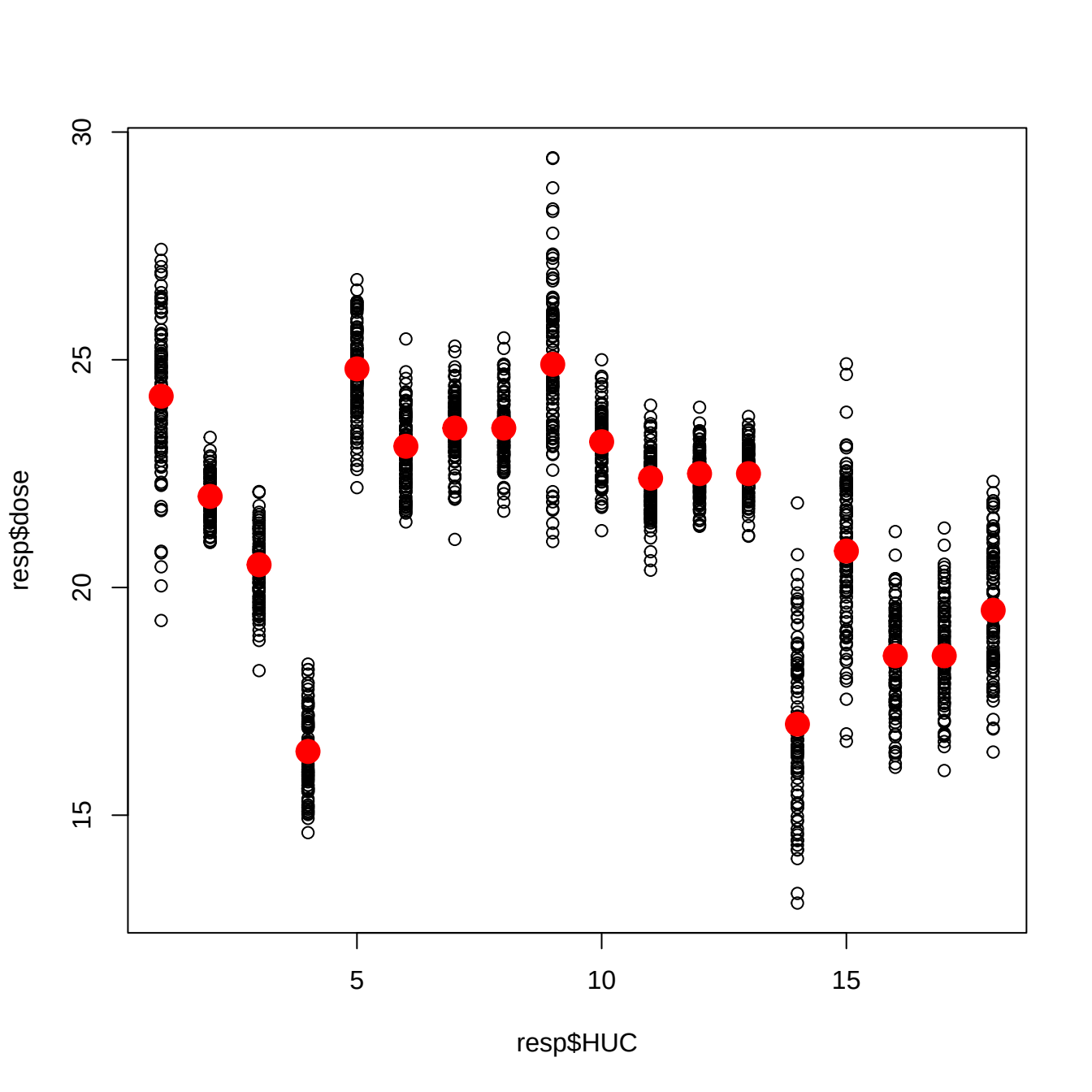
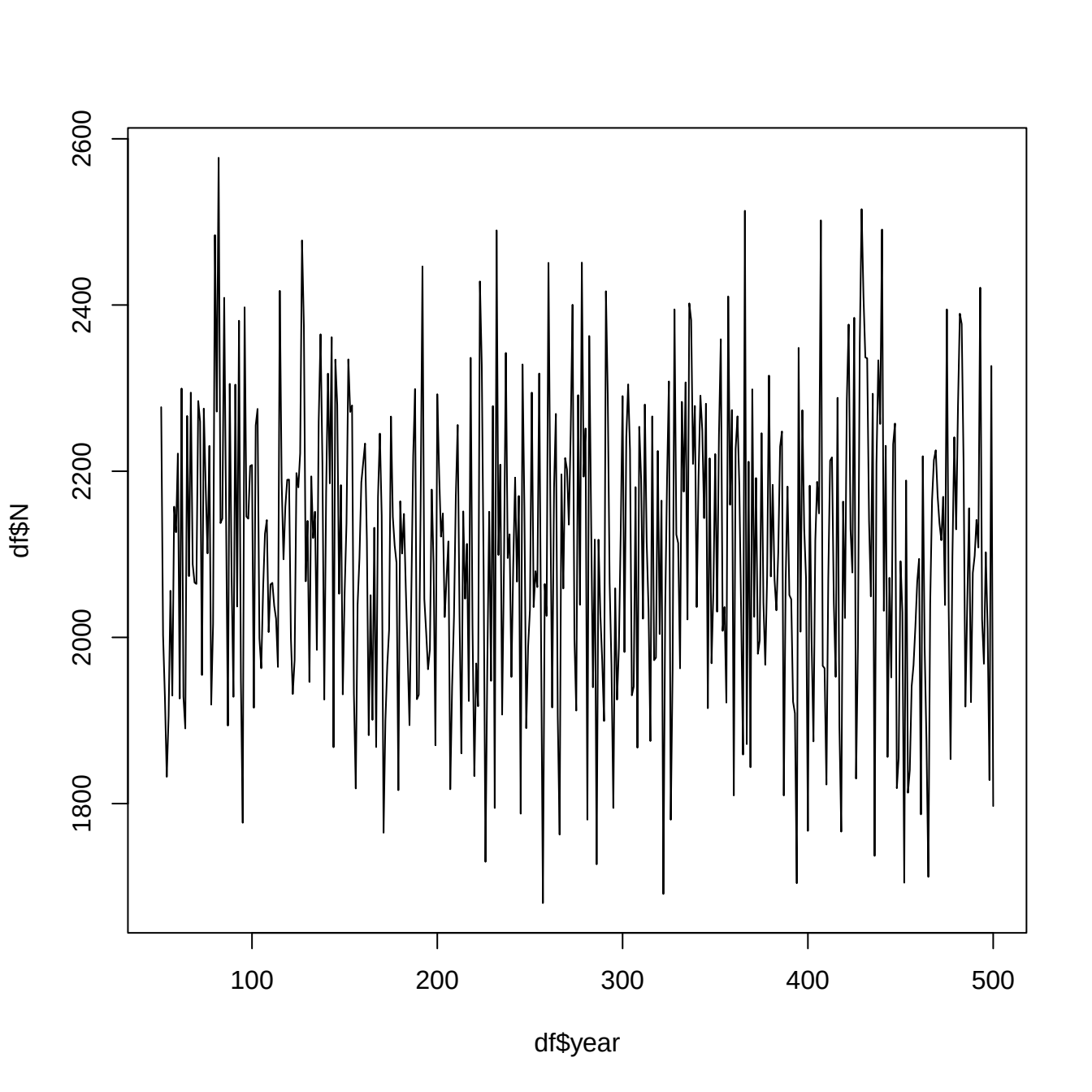
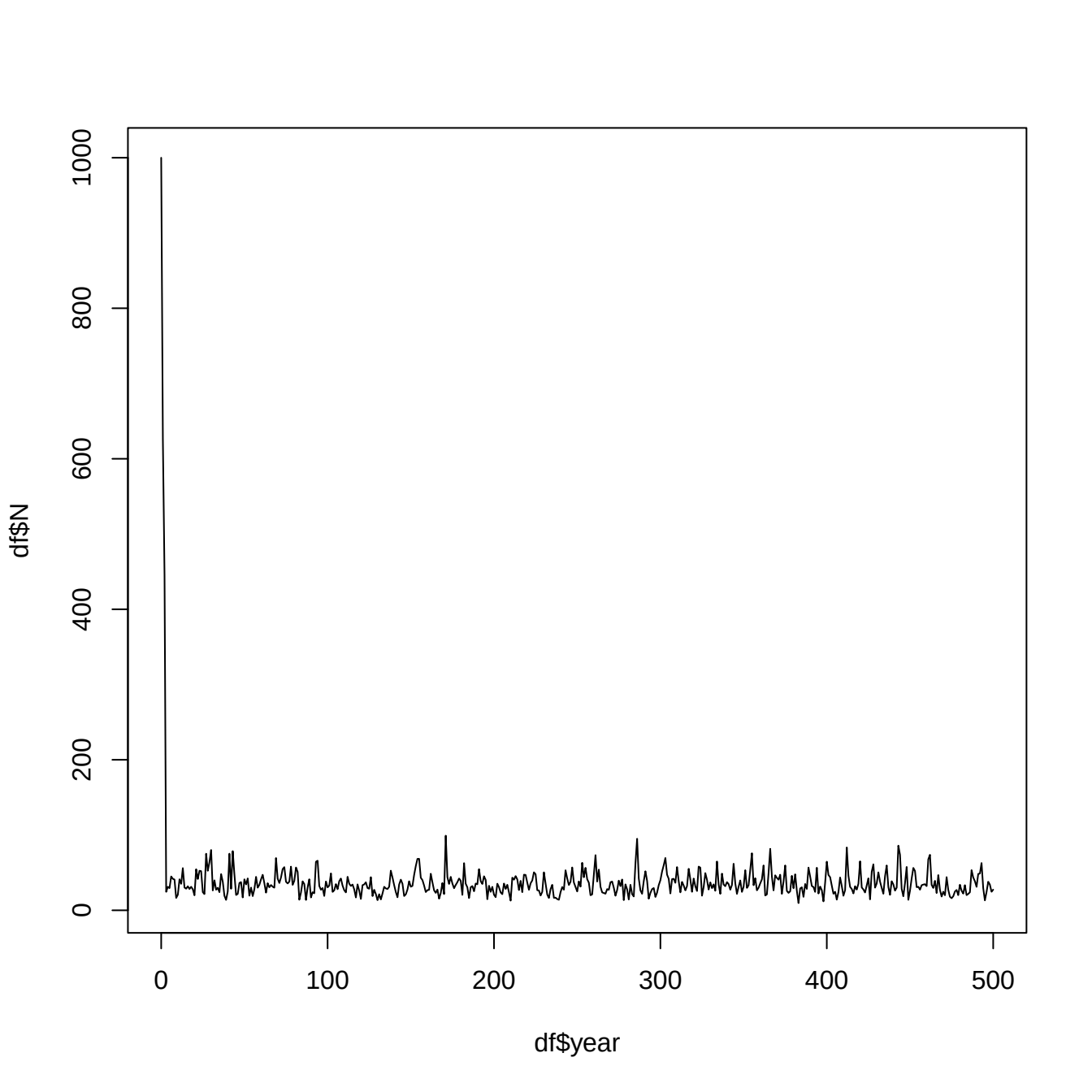
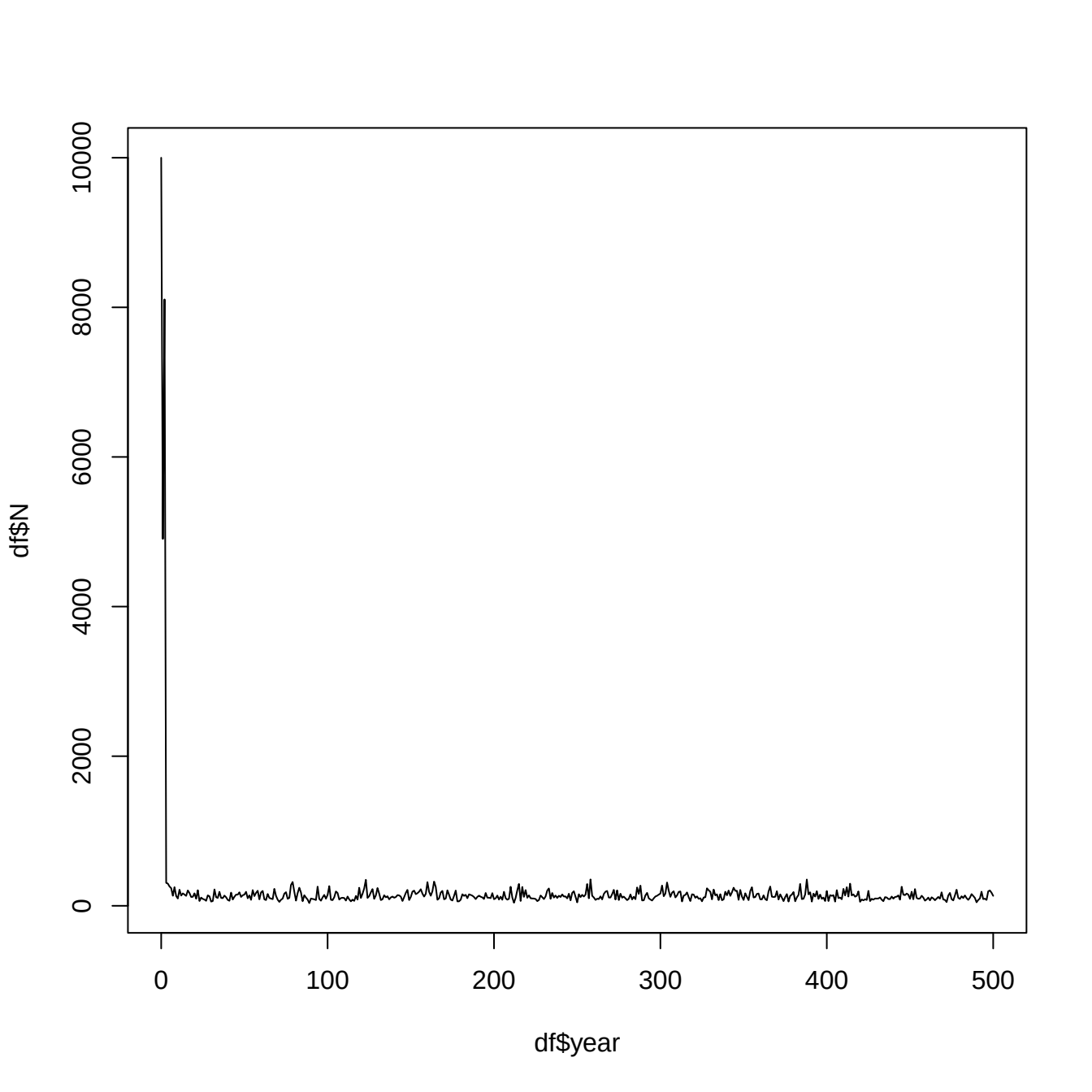
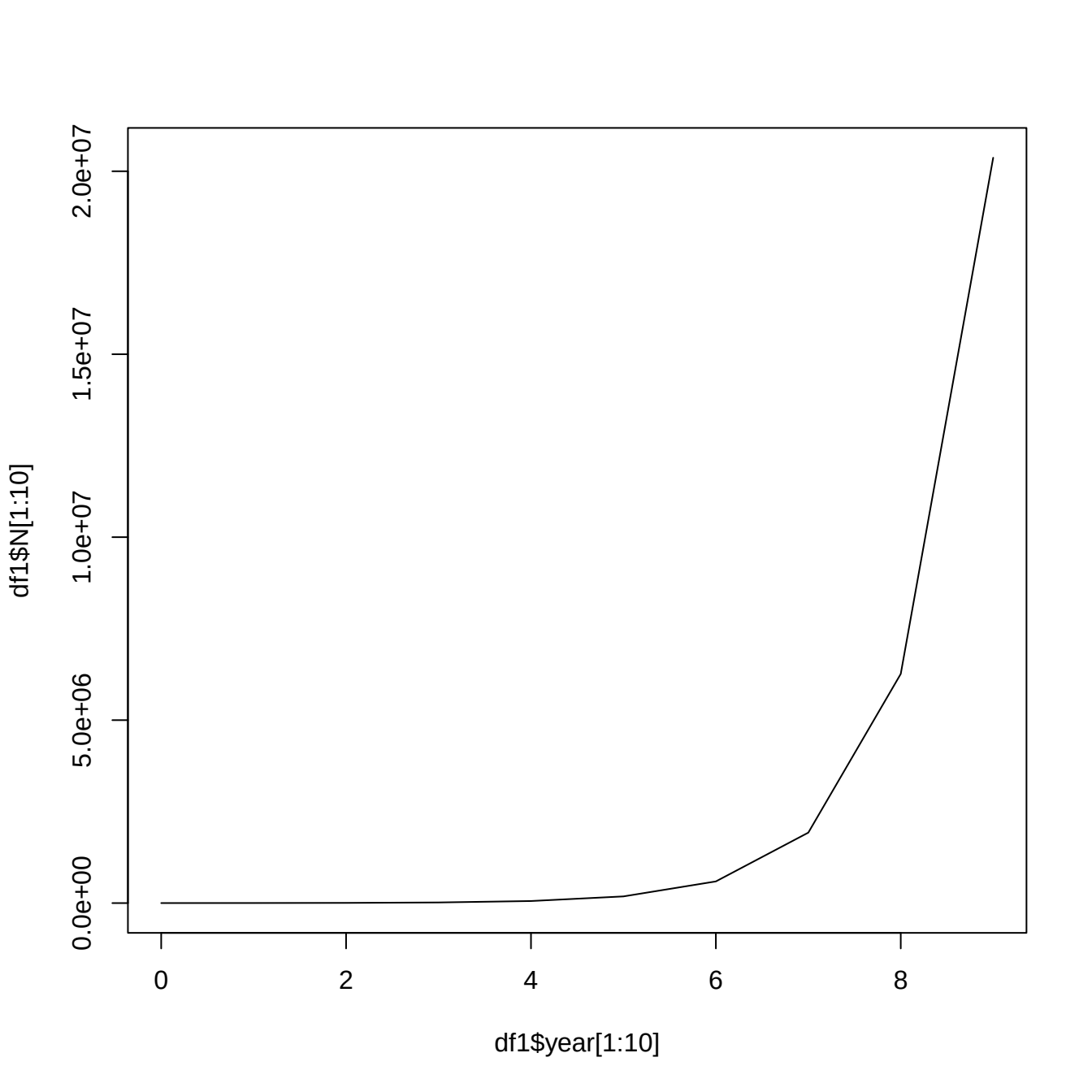
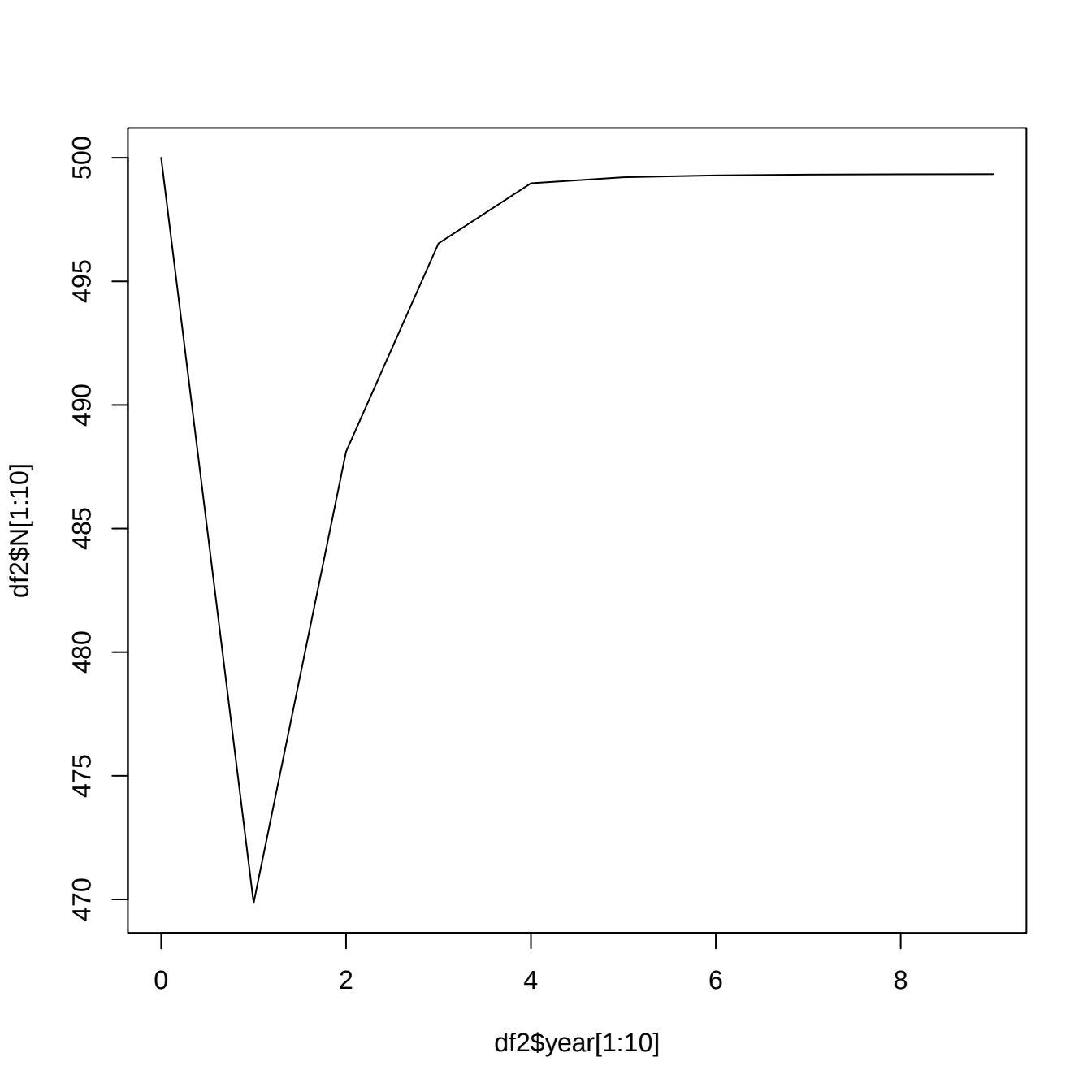

resp$dose
30
25
20
15
5
10
15
resp$HUC

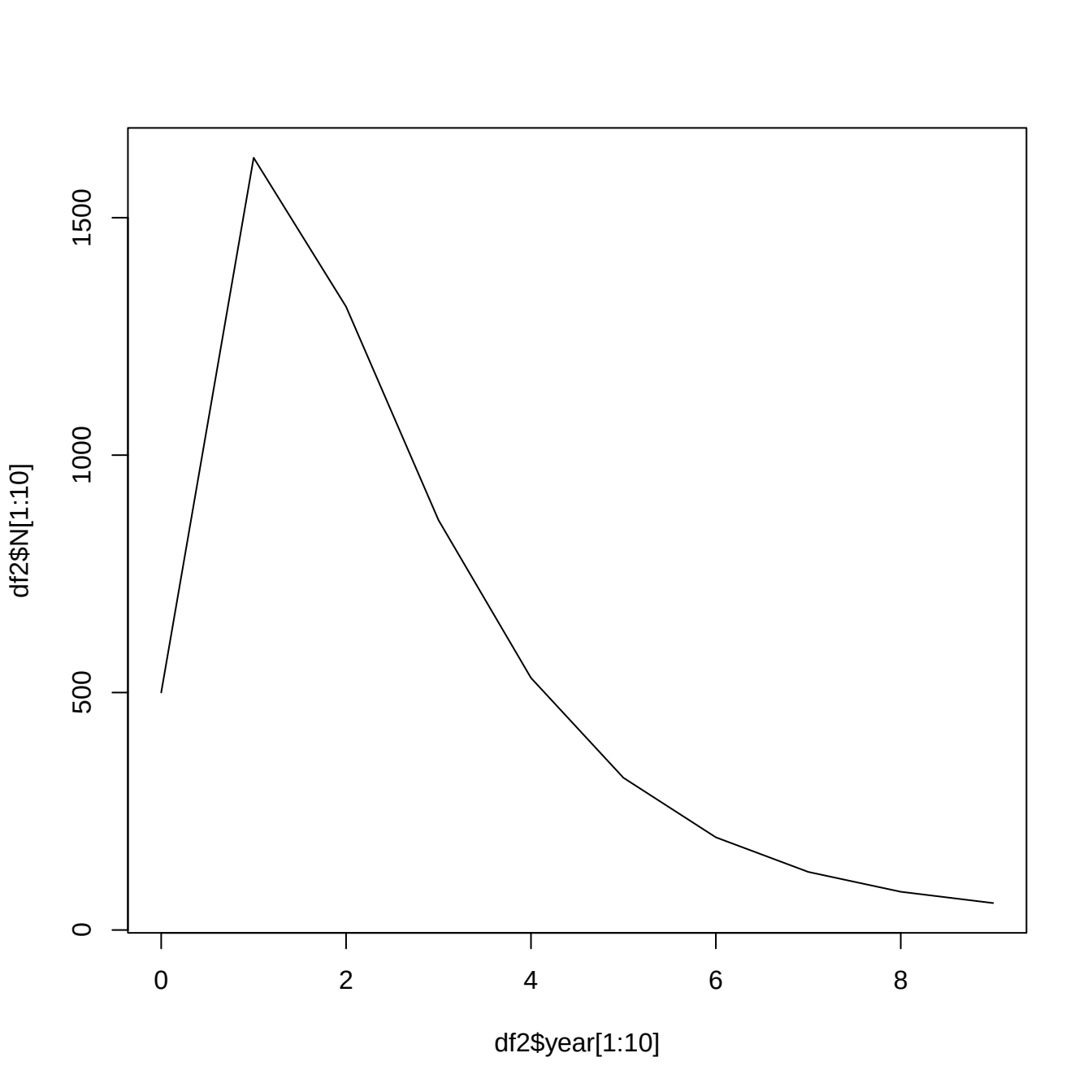

df$N
2600
2400
2200
2000
1800
100
200
300
400
500
df$year

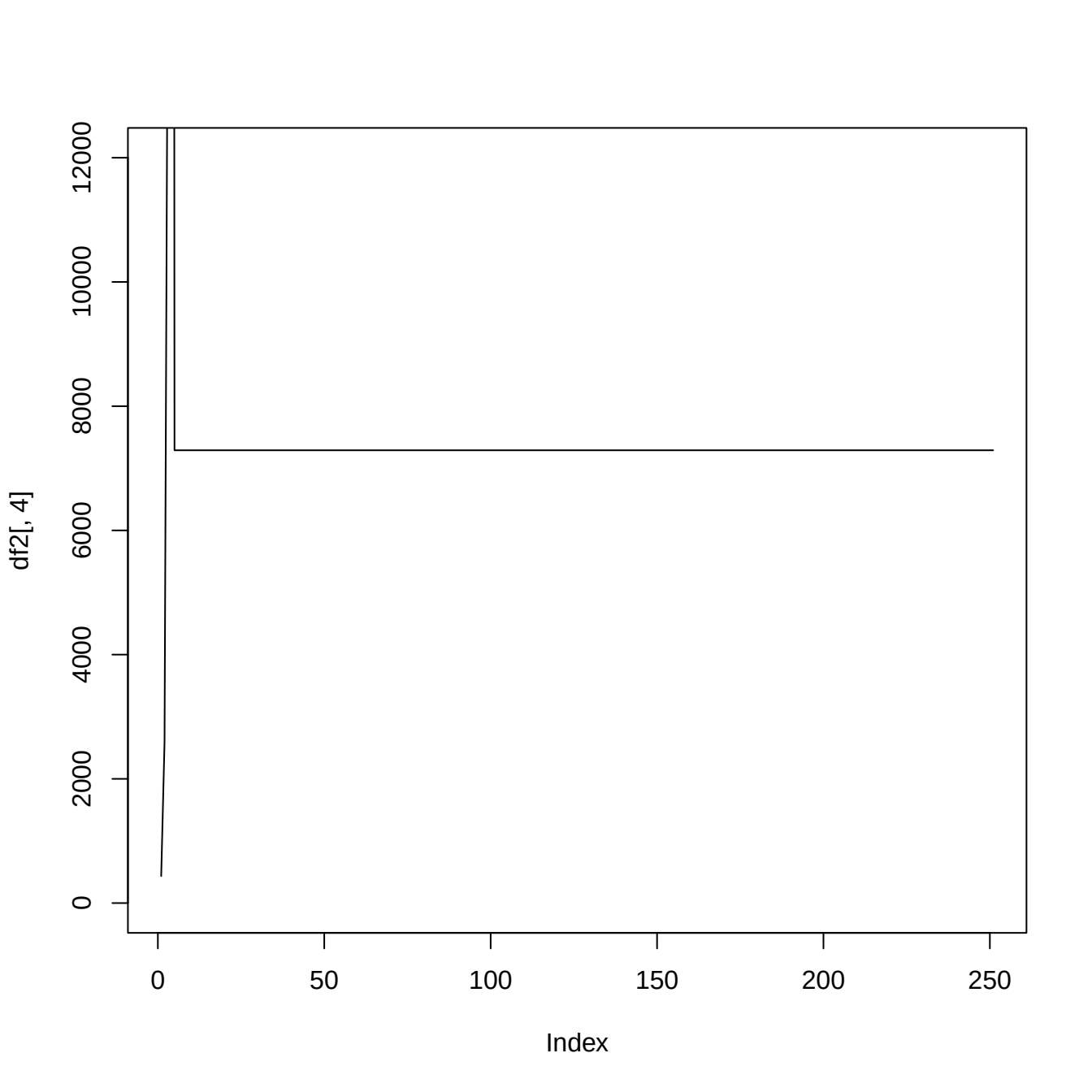

df$N
0
200
400
600
800
1000
0
100
200
300
400
500
df$year

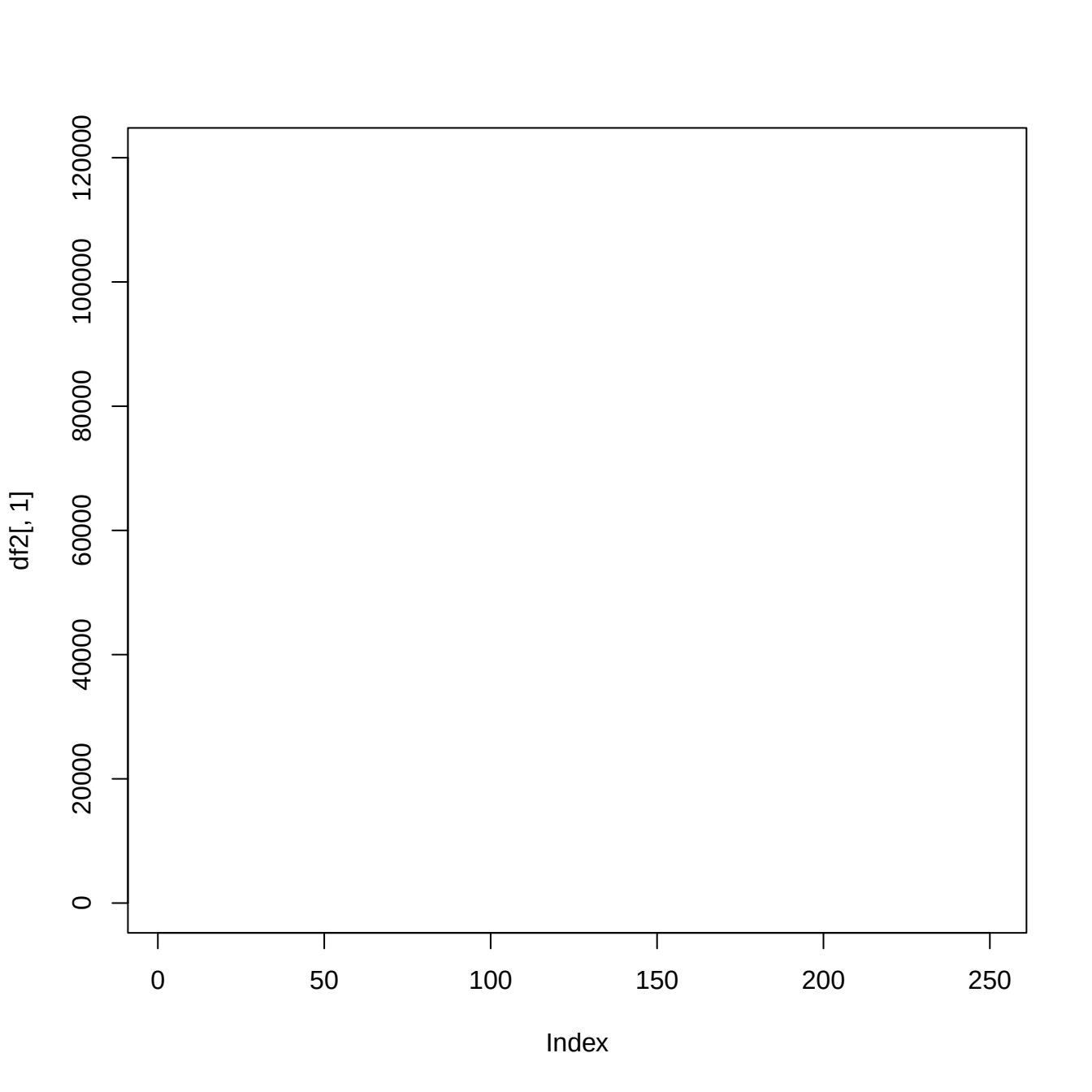

df$N
0
2000
4000
6000
8000
10000
0
100
200
300
400
500
df$year

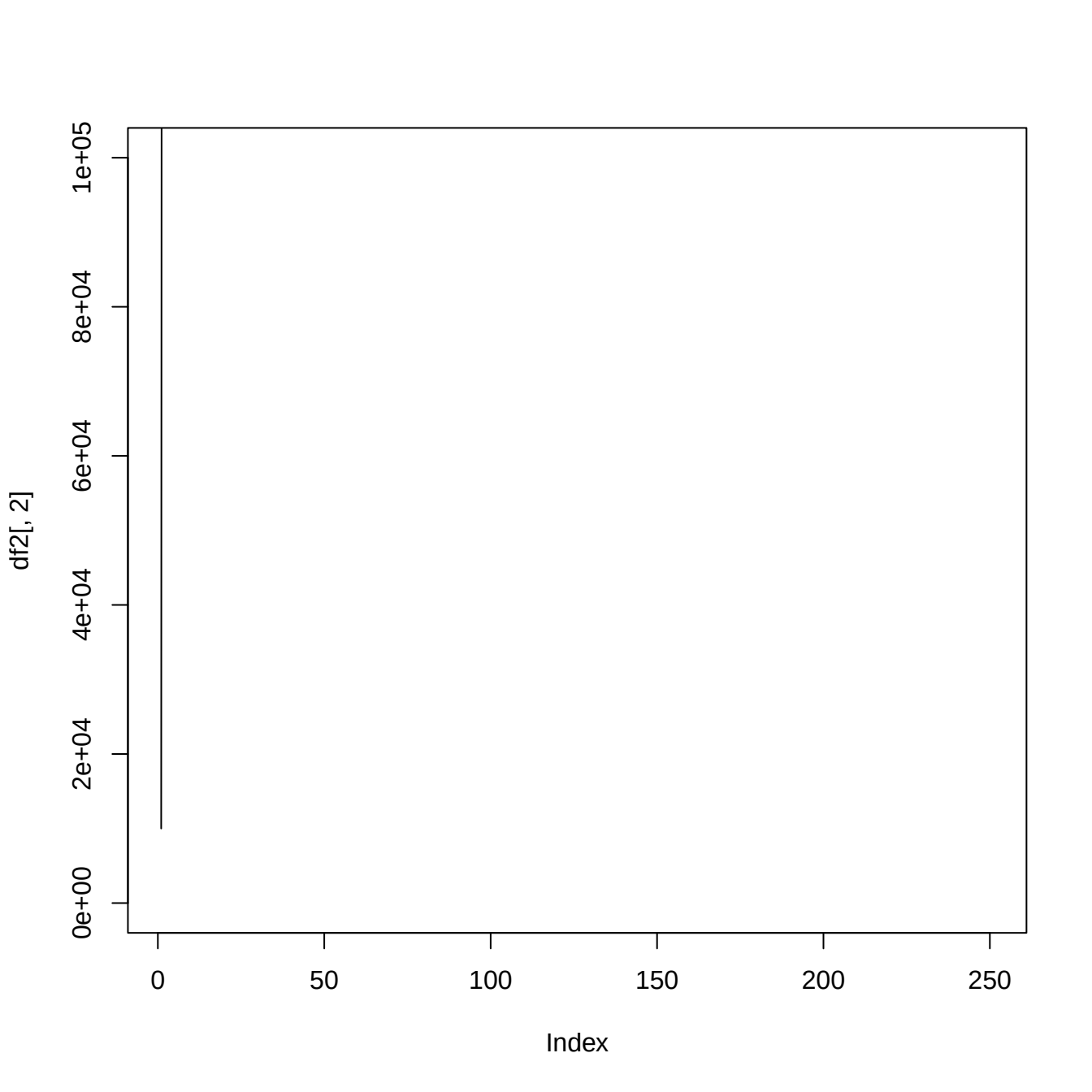

df1$N[1:10]
0.0e+00
5.0e+06
1.0e+07
1.5e+07
2.0e+07
0
2
4
6
8
df1$year[1:10]

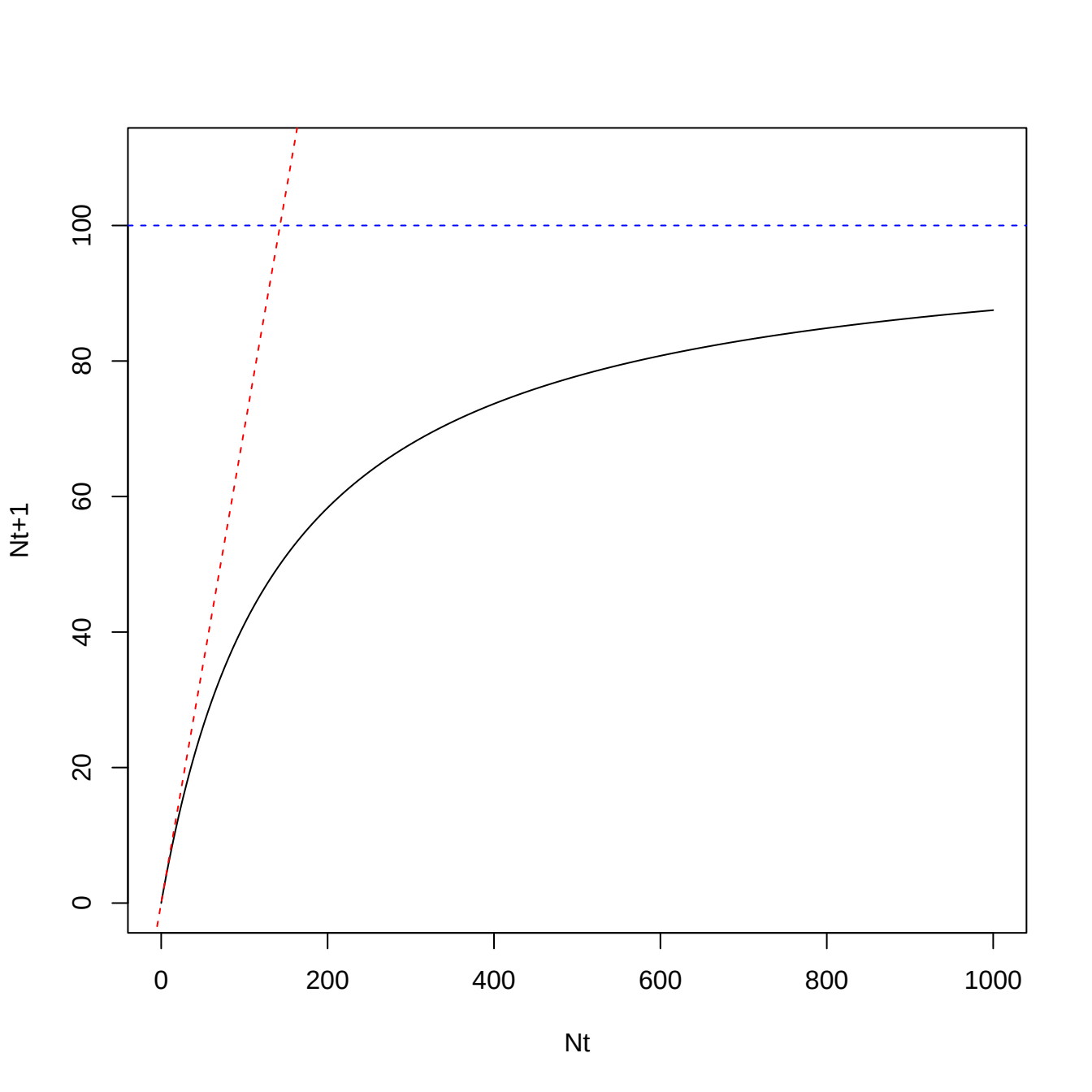

df2$N[1:10]
500
495
490
485
480
475
470
0
2
4
6
8
df2$year[1:10]

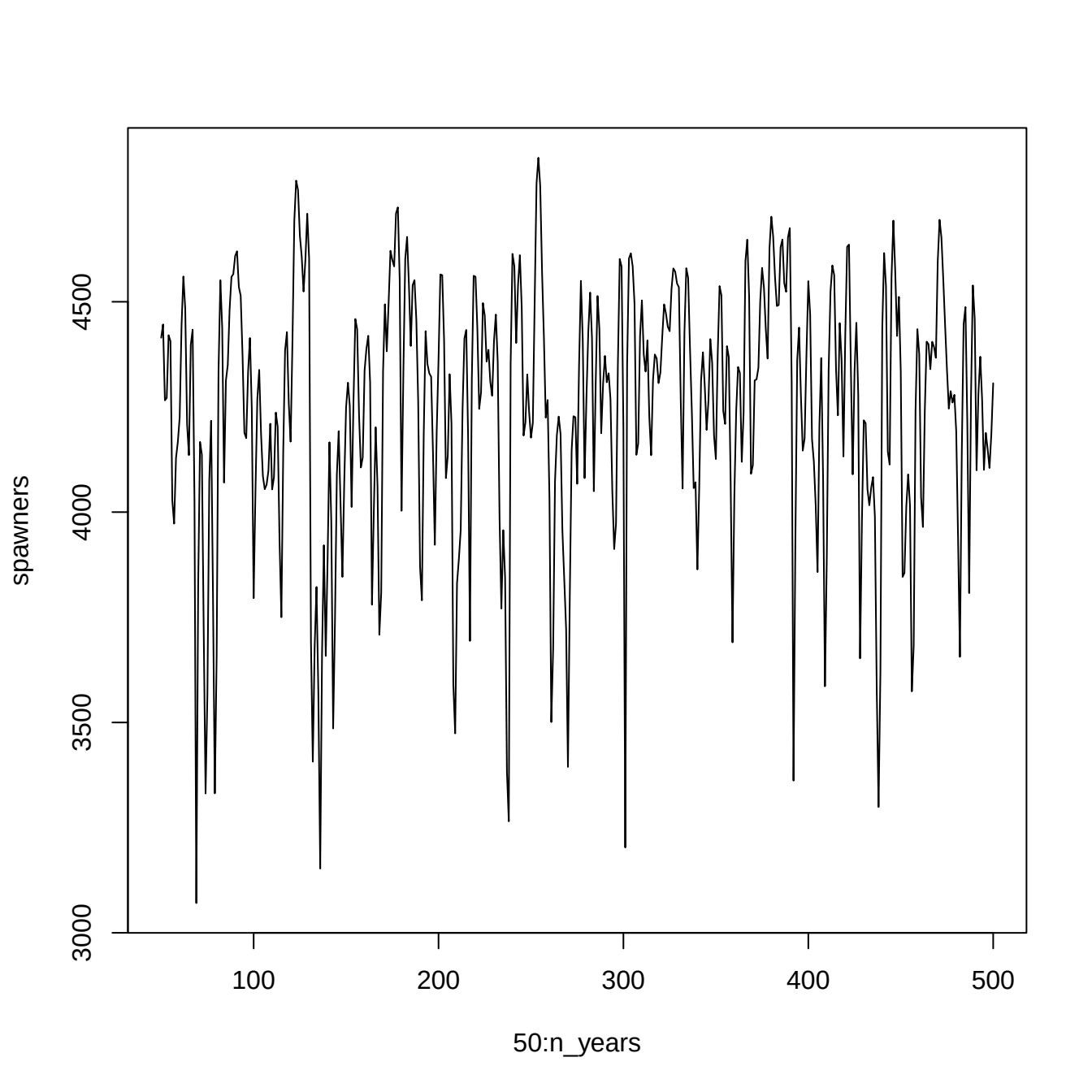

df2$N[1:10]
0
500
1000
1500
0
2
4
6
8
df2$year[1:10]

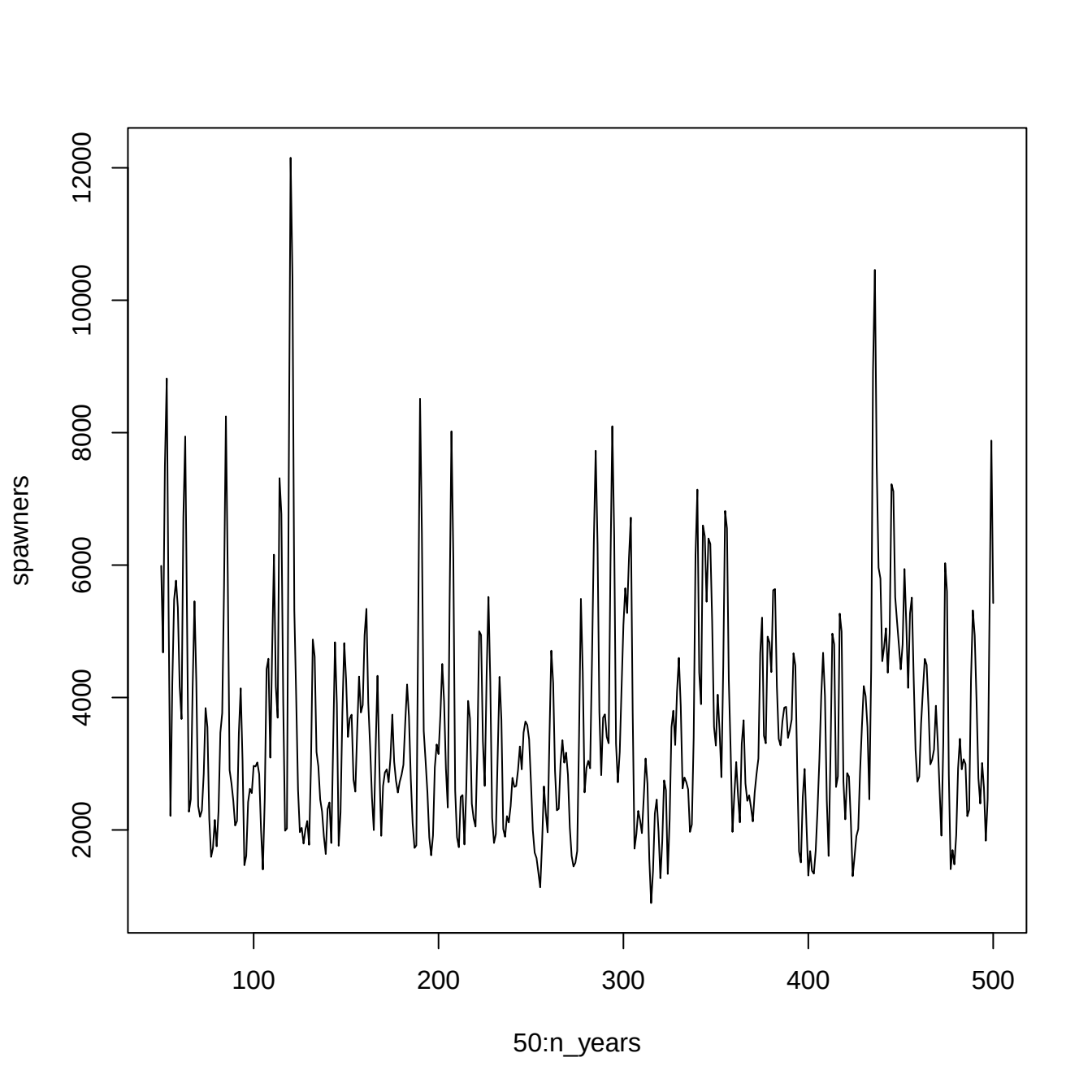

df2[, 4]
0
2000
4000
6000
8000
10000
12000
0
50
100
150
200
250
Index

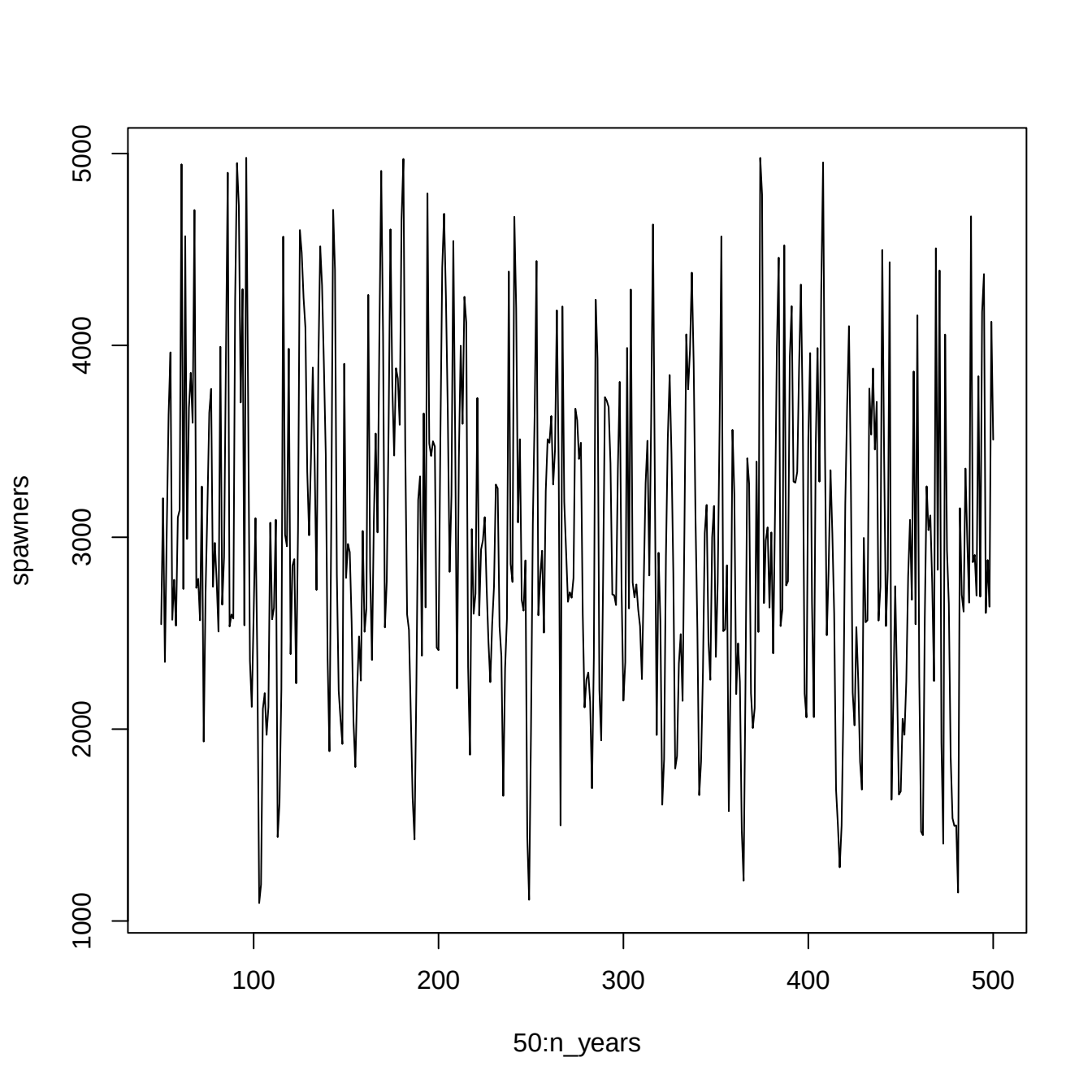

df2[, 1]
0
20000
40000
60000
80000
100000
120000
0
50
100
150
200
250
Index

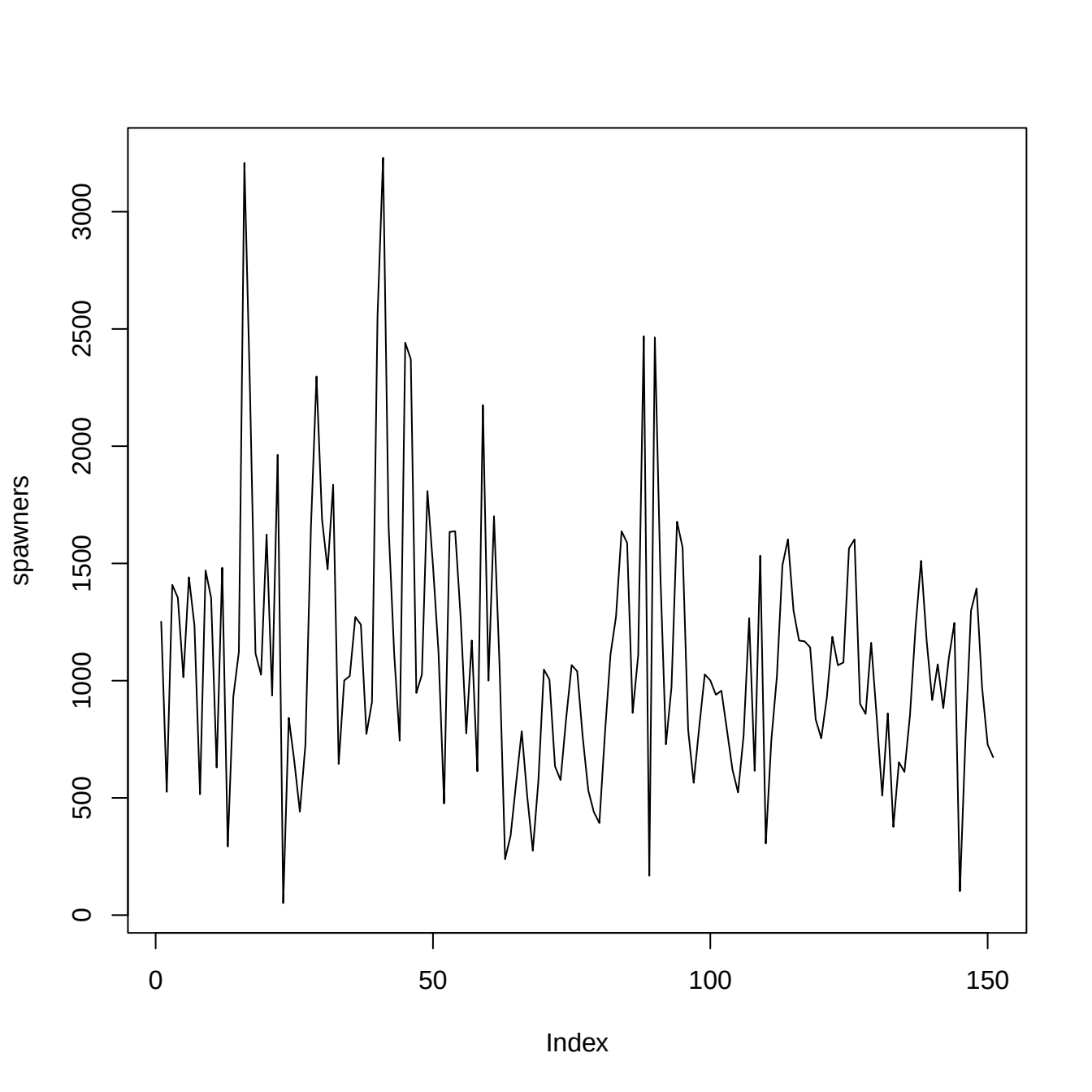

df2[, 2]
0e+00
2e+04
4e+04
6e+04
8e+04
1e+05
0
50
100
150
200
250
Index

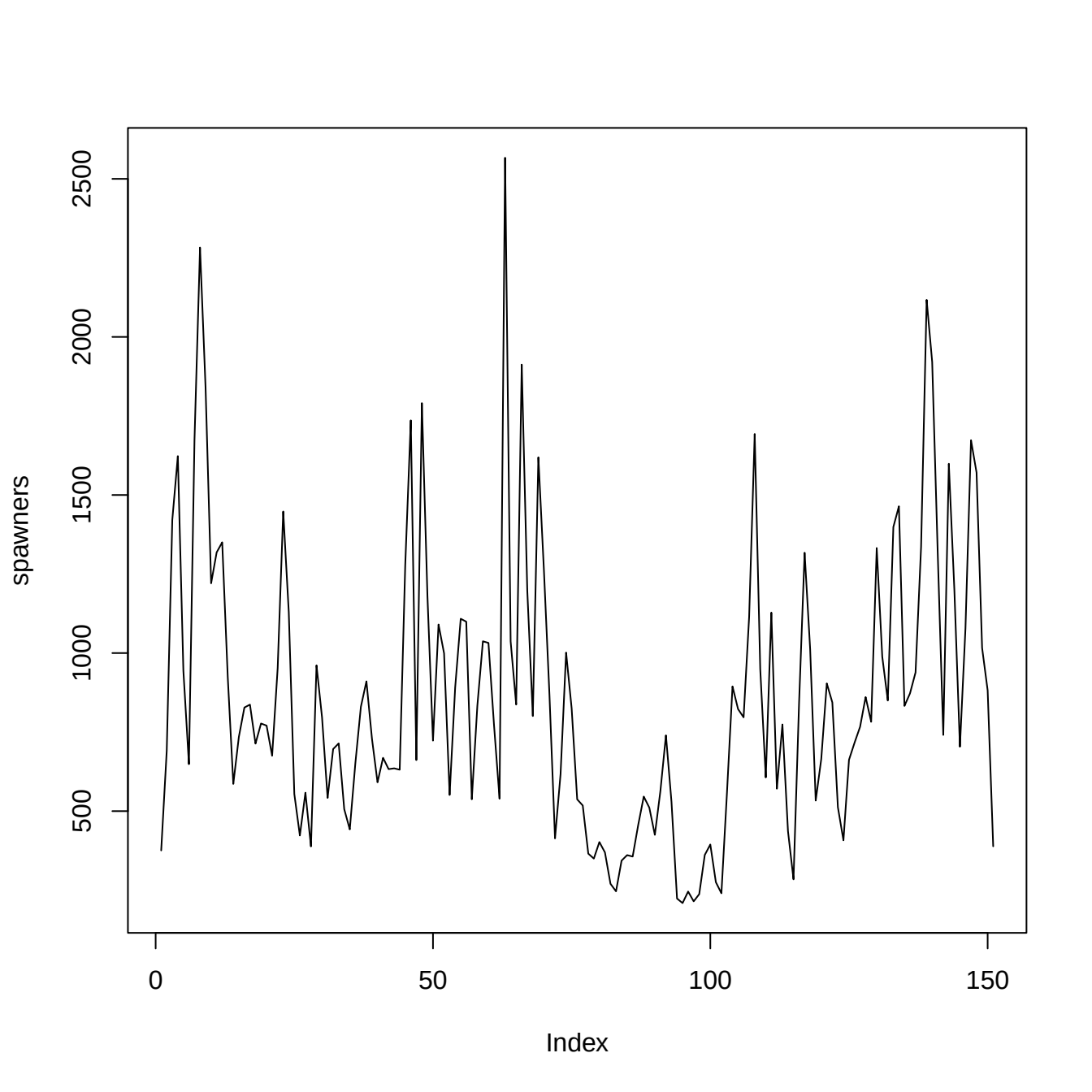

Nt+1
0
20
40
60
80
100
0
200
400
600
800
1000
Nt

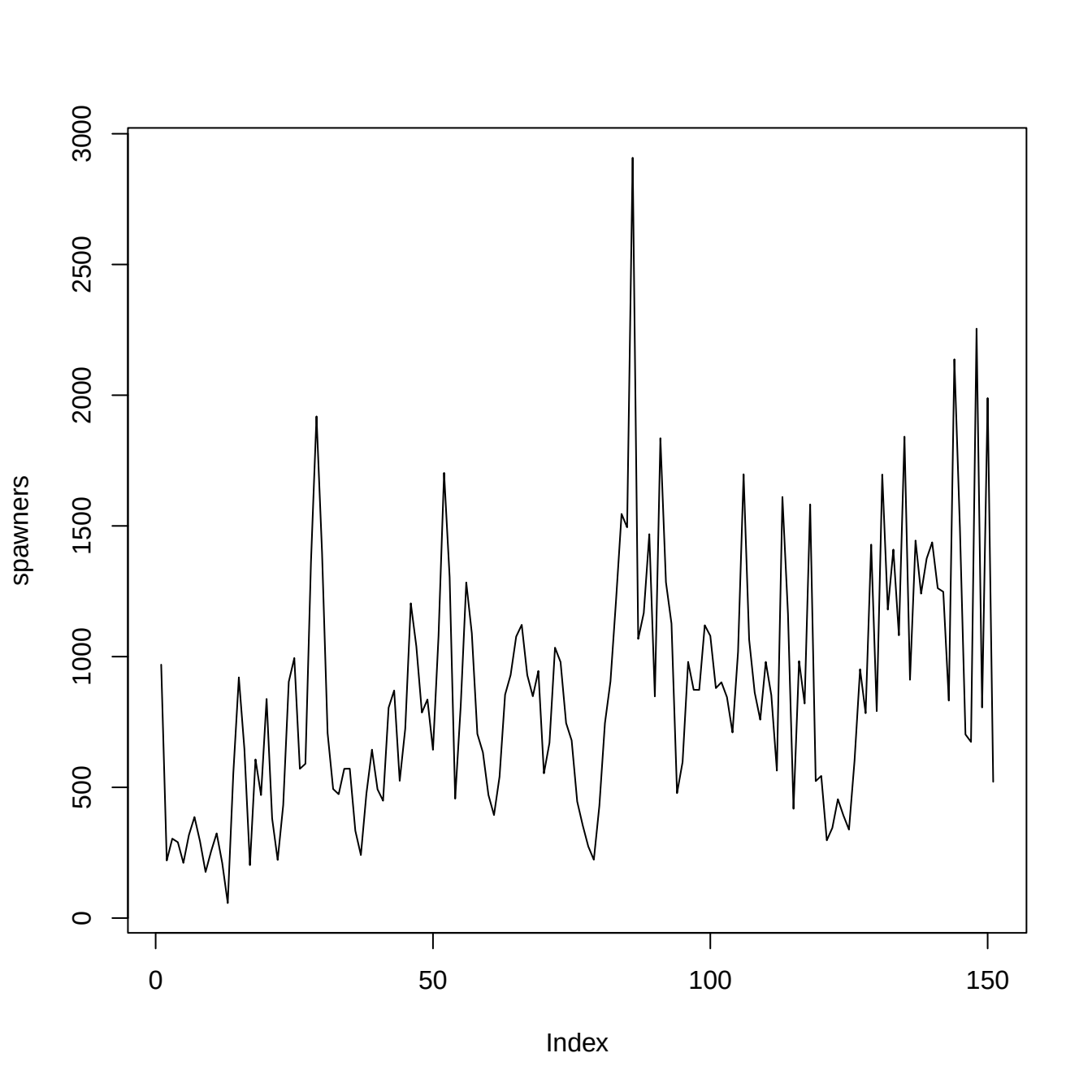

spawners
3000
3500
4000
4500
100
200
300
400
500
50:n_years

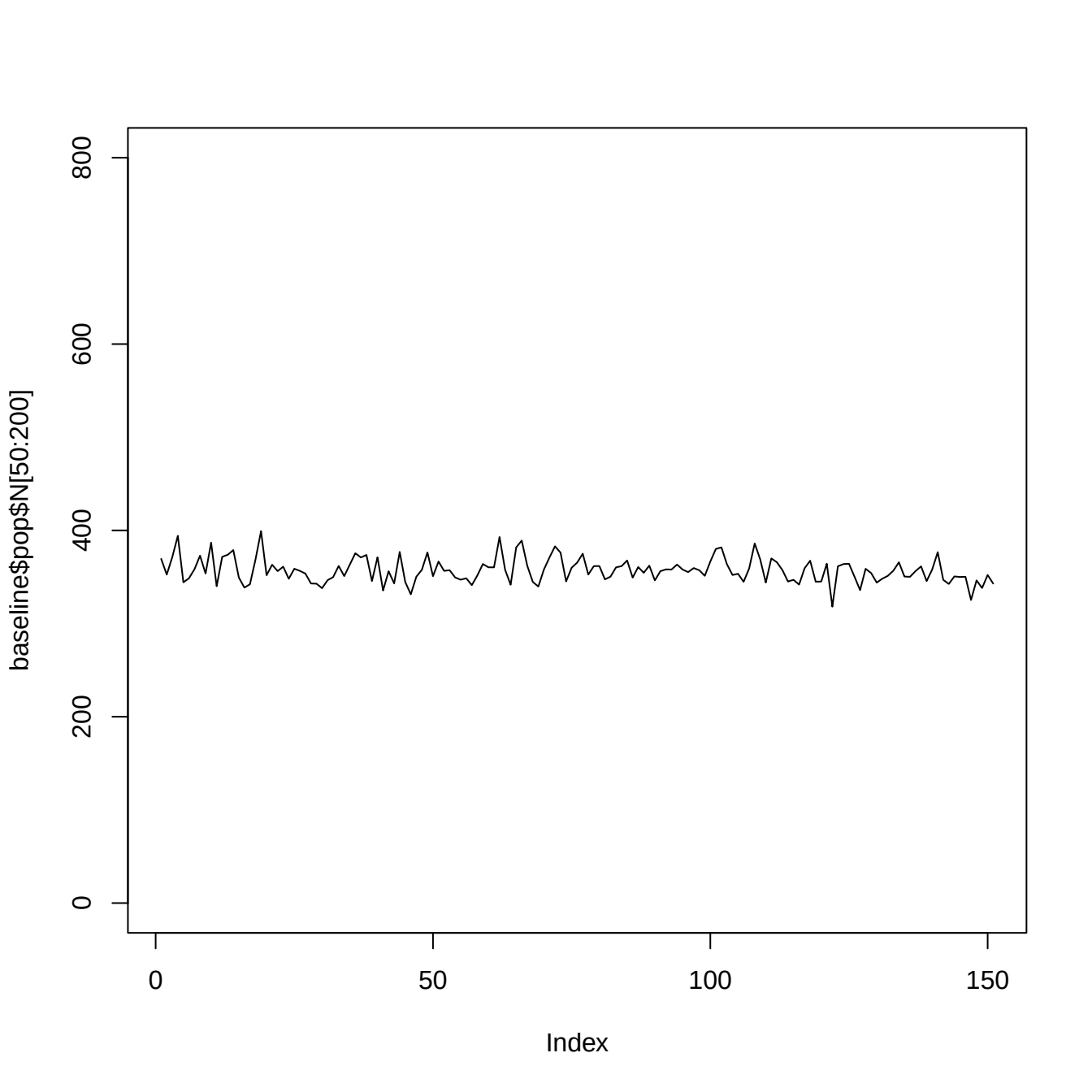

spawners
2000
4000
6000
8000
10000
12000
100
200
300
400
500
50:n_years

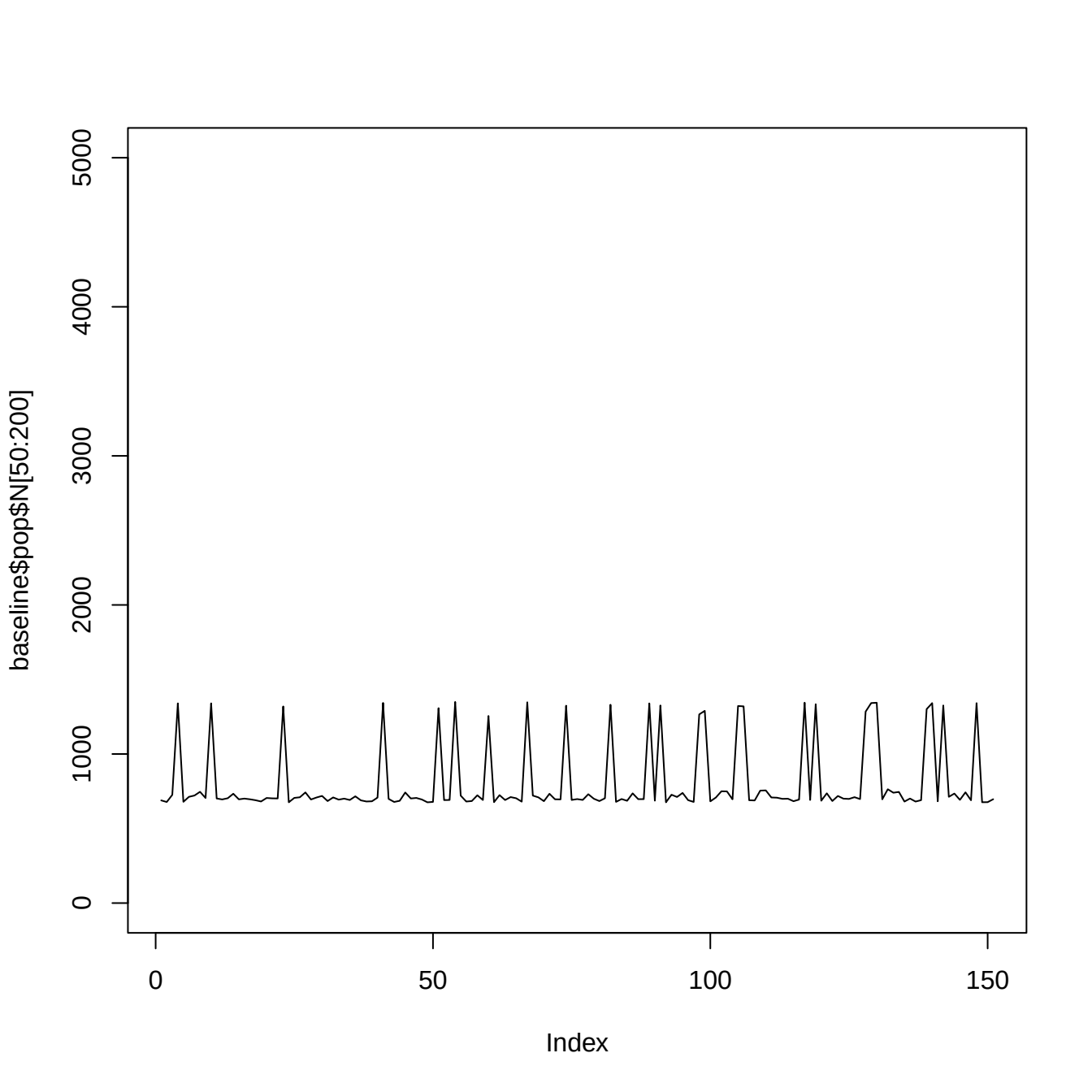

spawners
1000
2000
3000
4000
5000
100
200
300
400
500
50:n_years

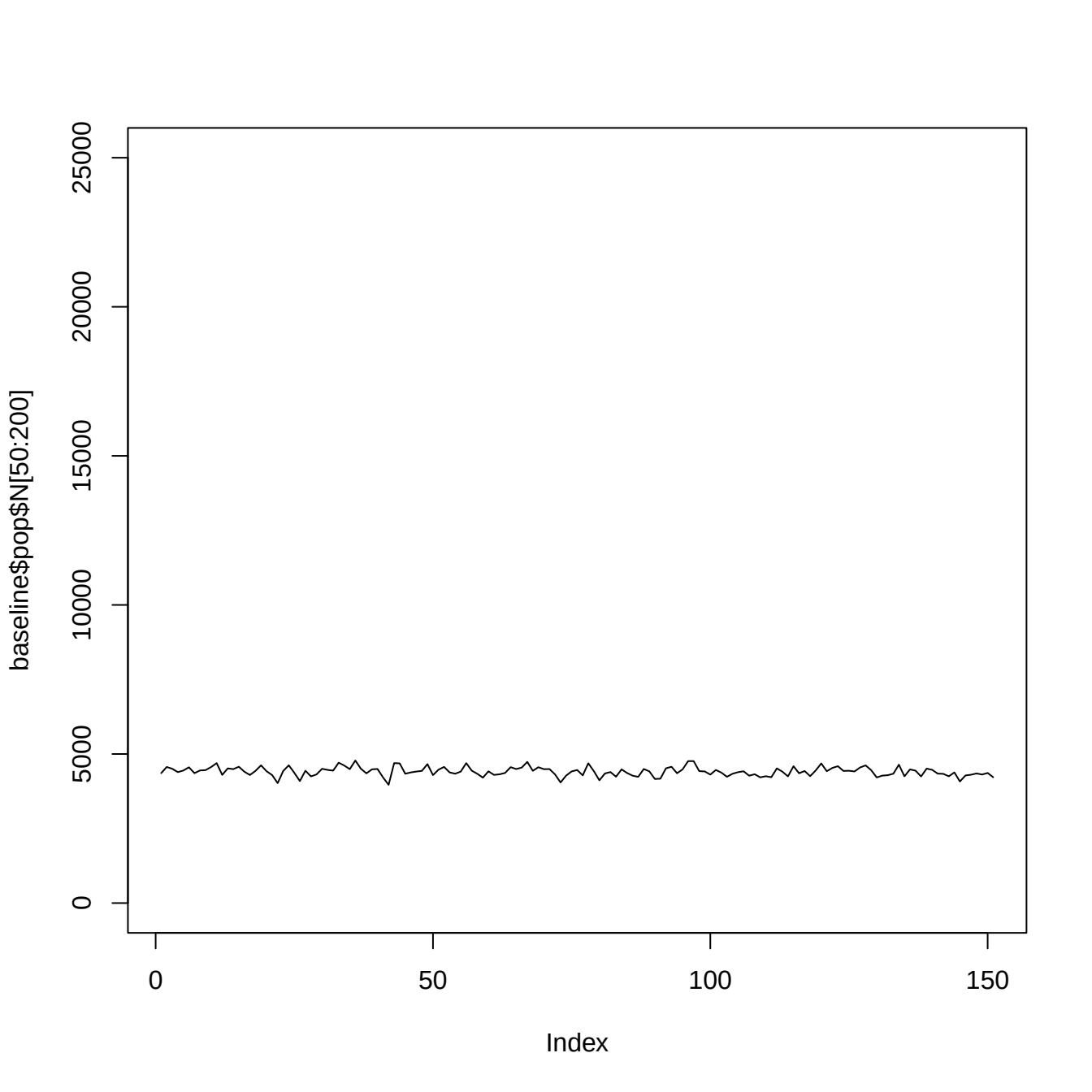

spawners
0
500
1000
1500
2000
2500
3000
0
50
100
150
Index

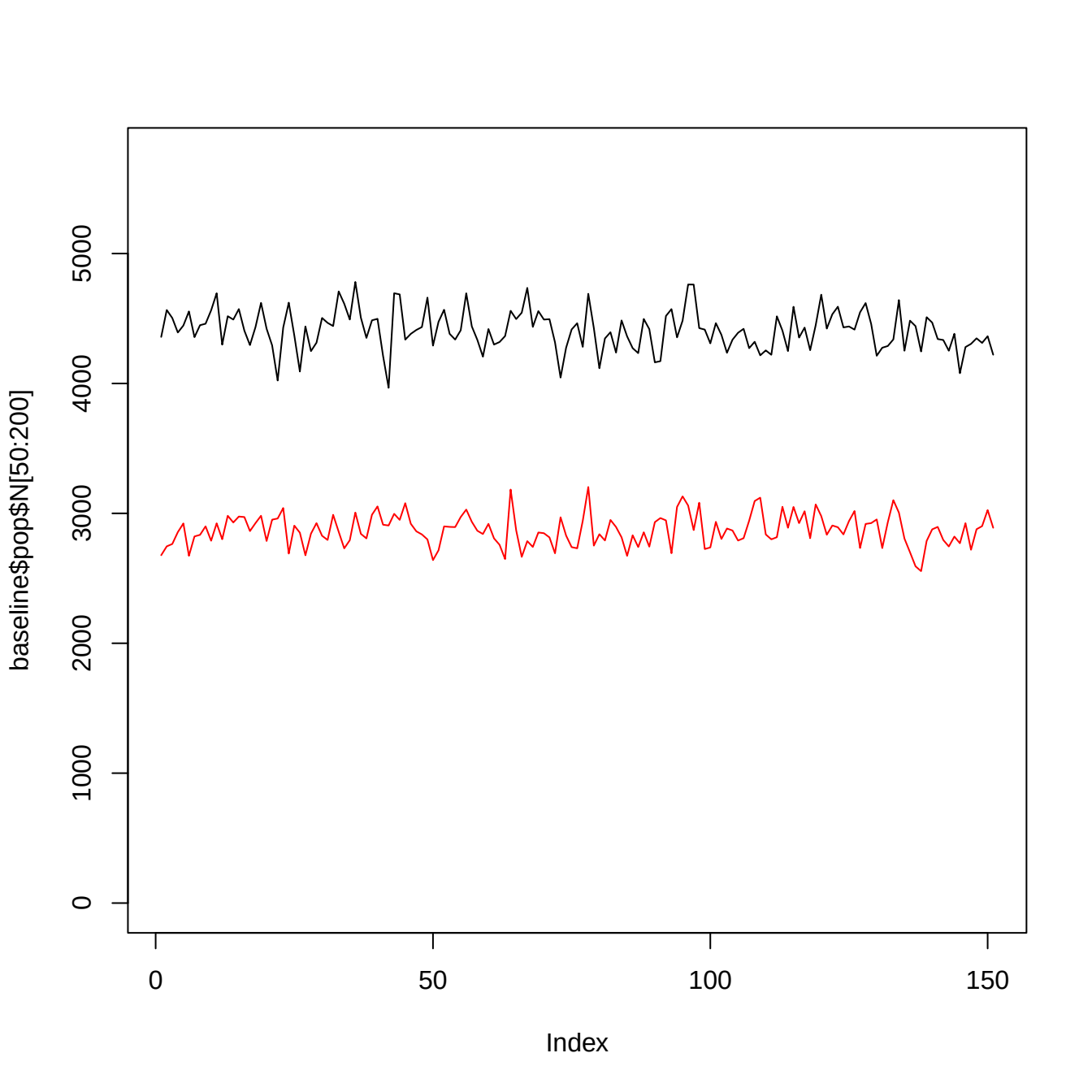

spawners
500
1000
1500
2000
2500
0
50
100
150
Index

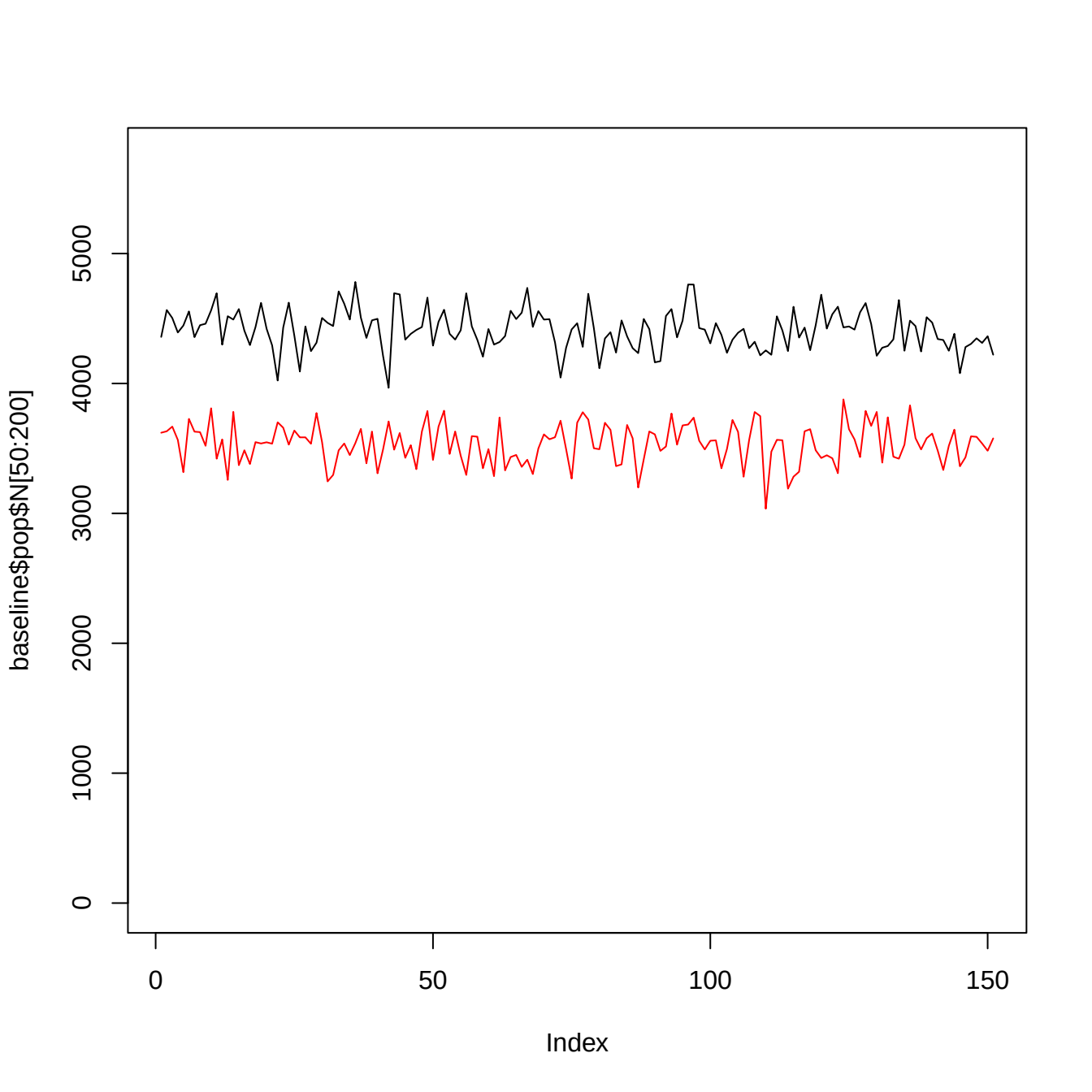

spawners
0
500
1000
1500
2000
2500
3000
0
50
100
150
Index

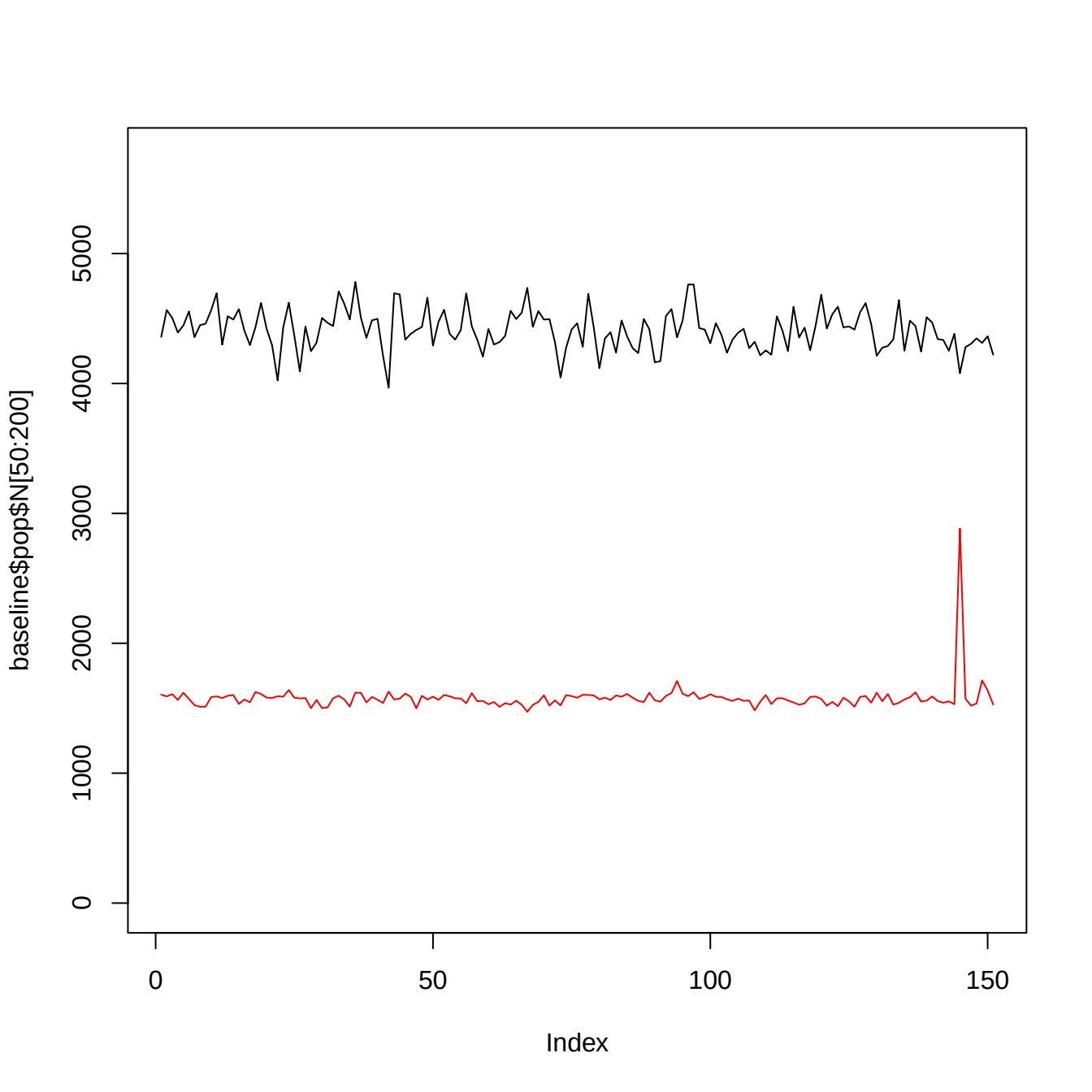

baselinepopN[50:200]
0
200
400
600
800
0
50
100
150
Index

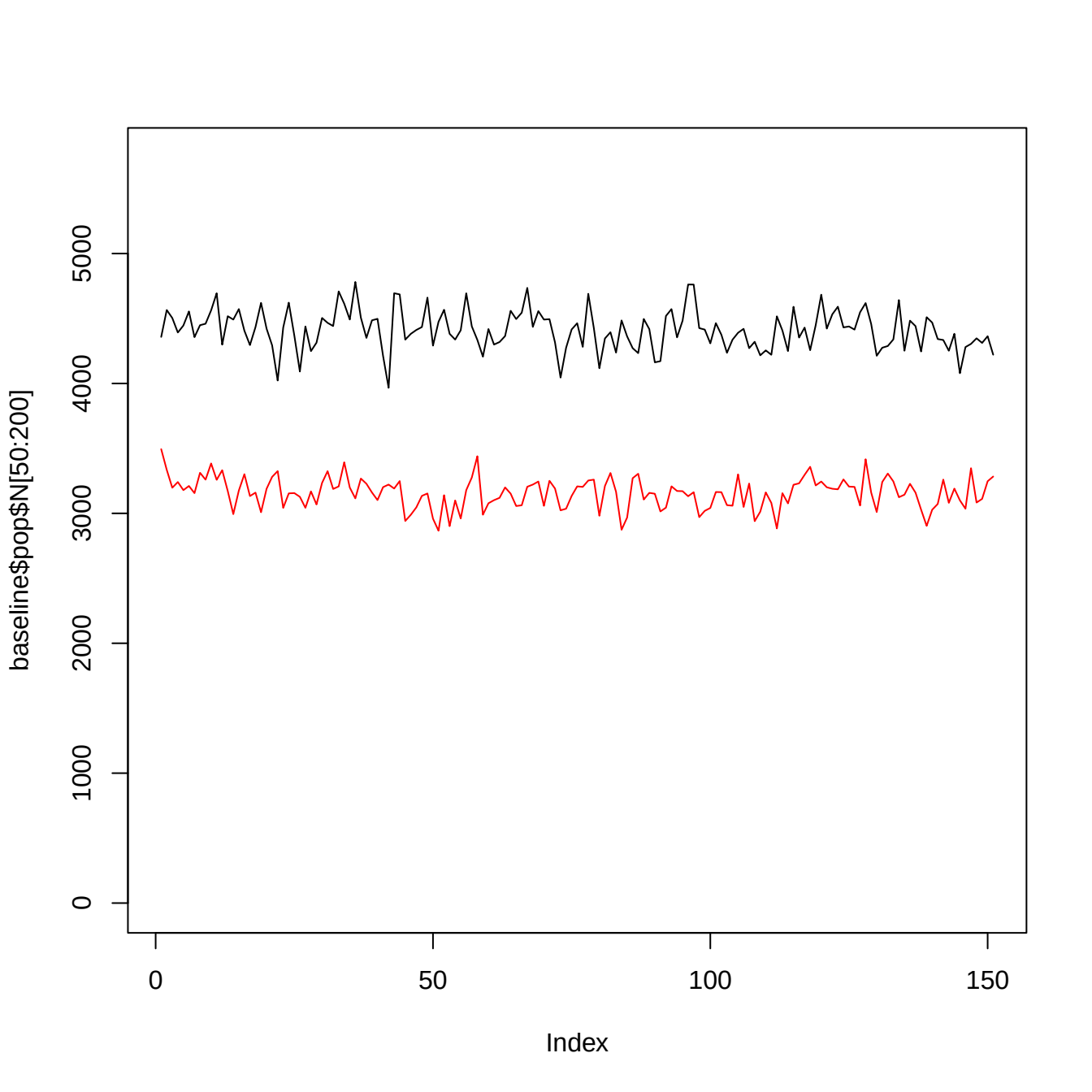

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

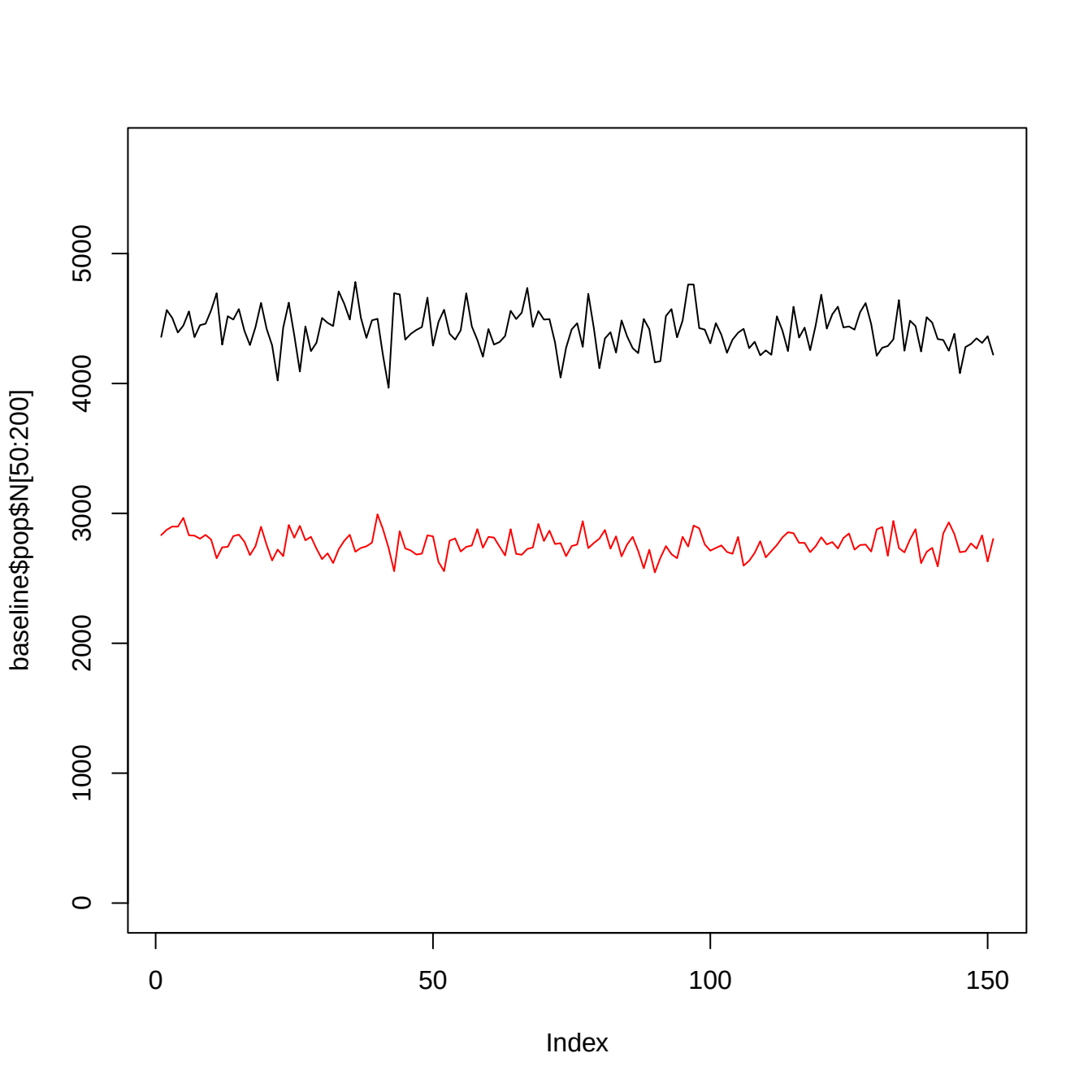

baselinepopN[50:200]
0
5000
10000
15000
20000
25000
0
50
100
150
Index

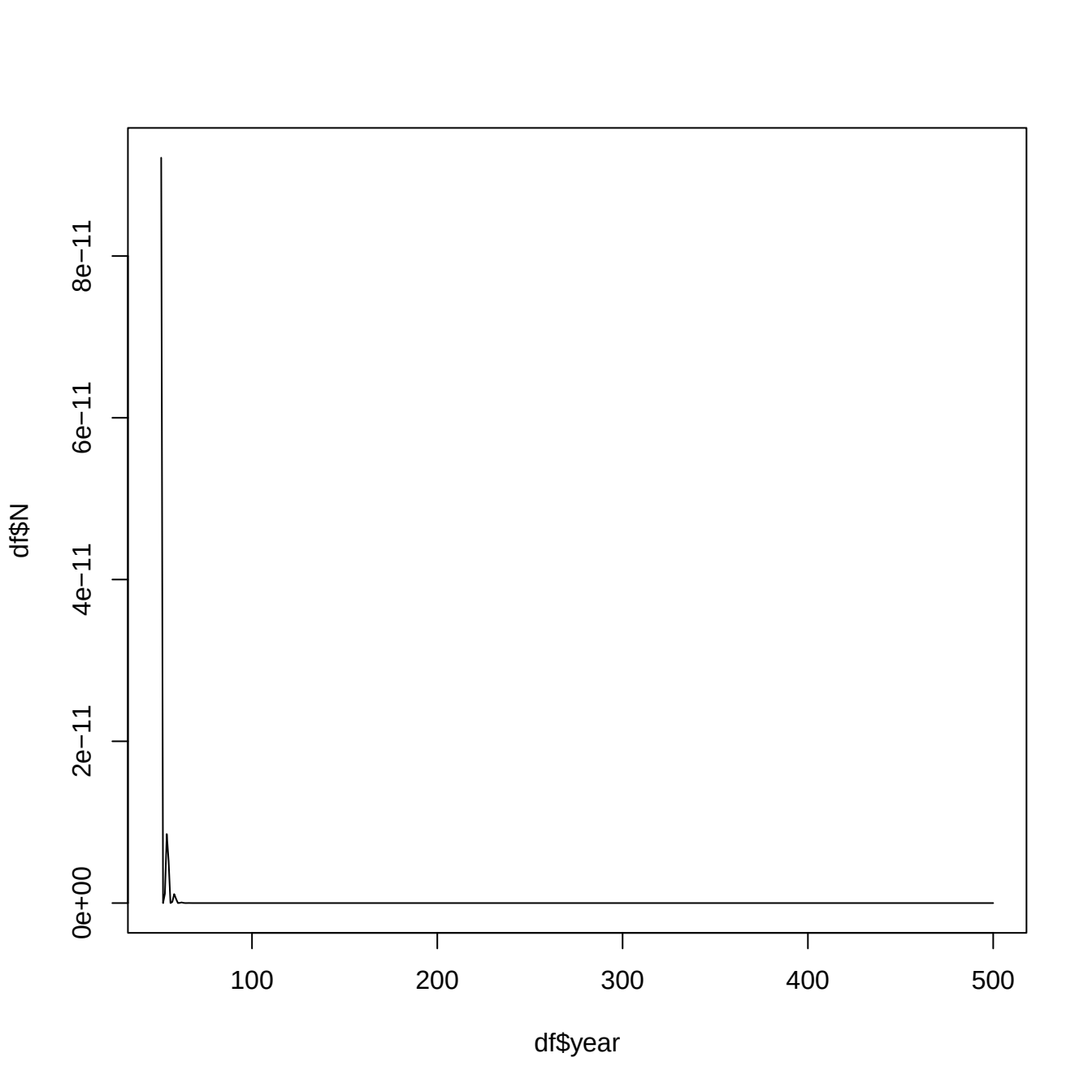

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

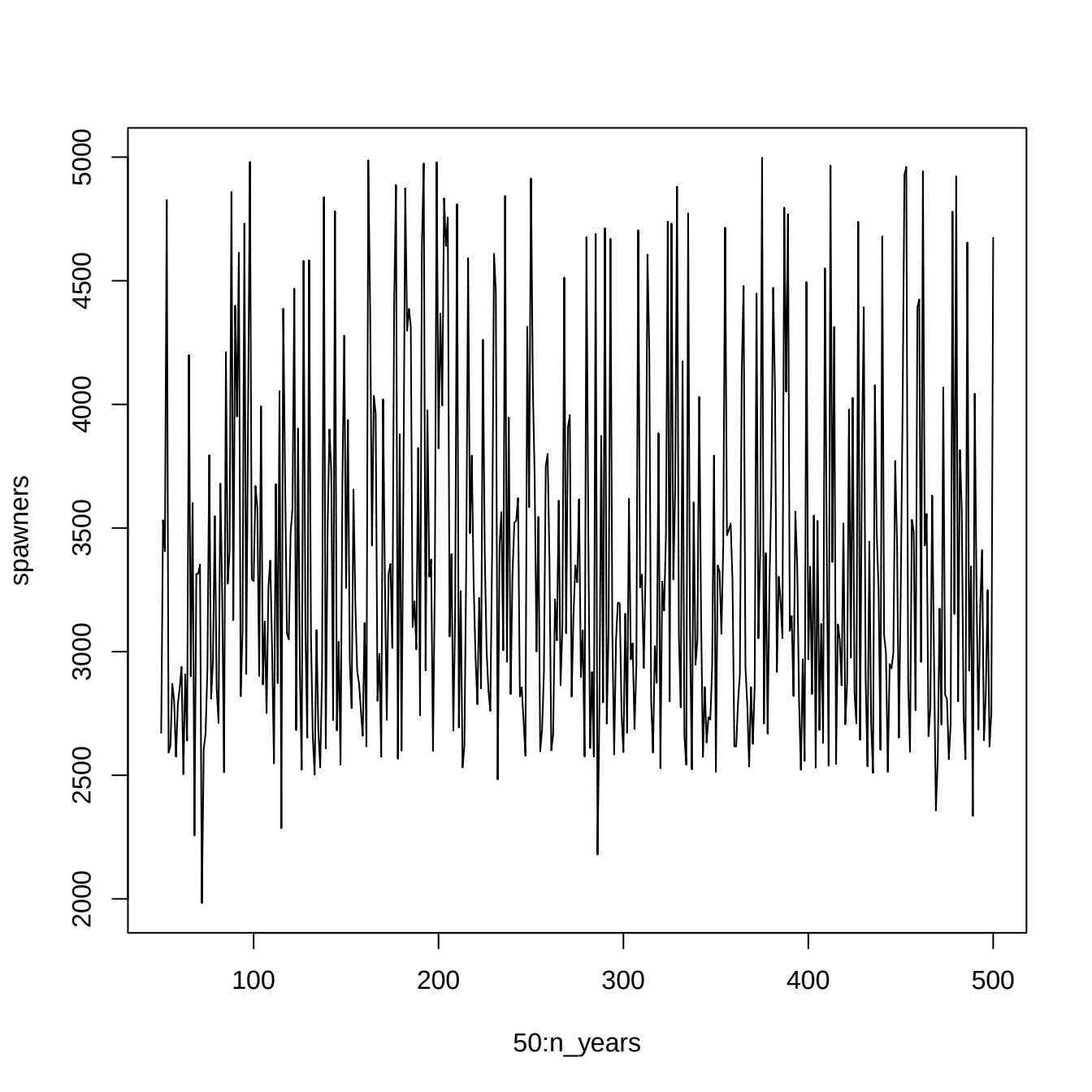

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

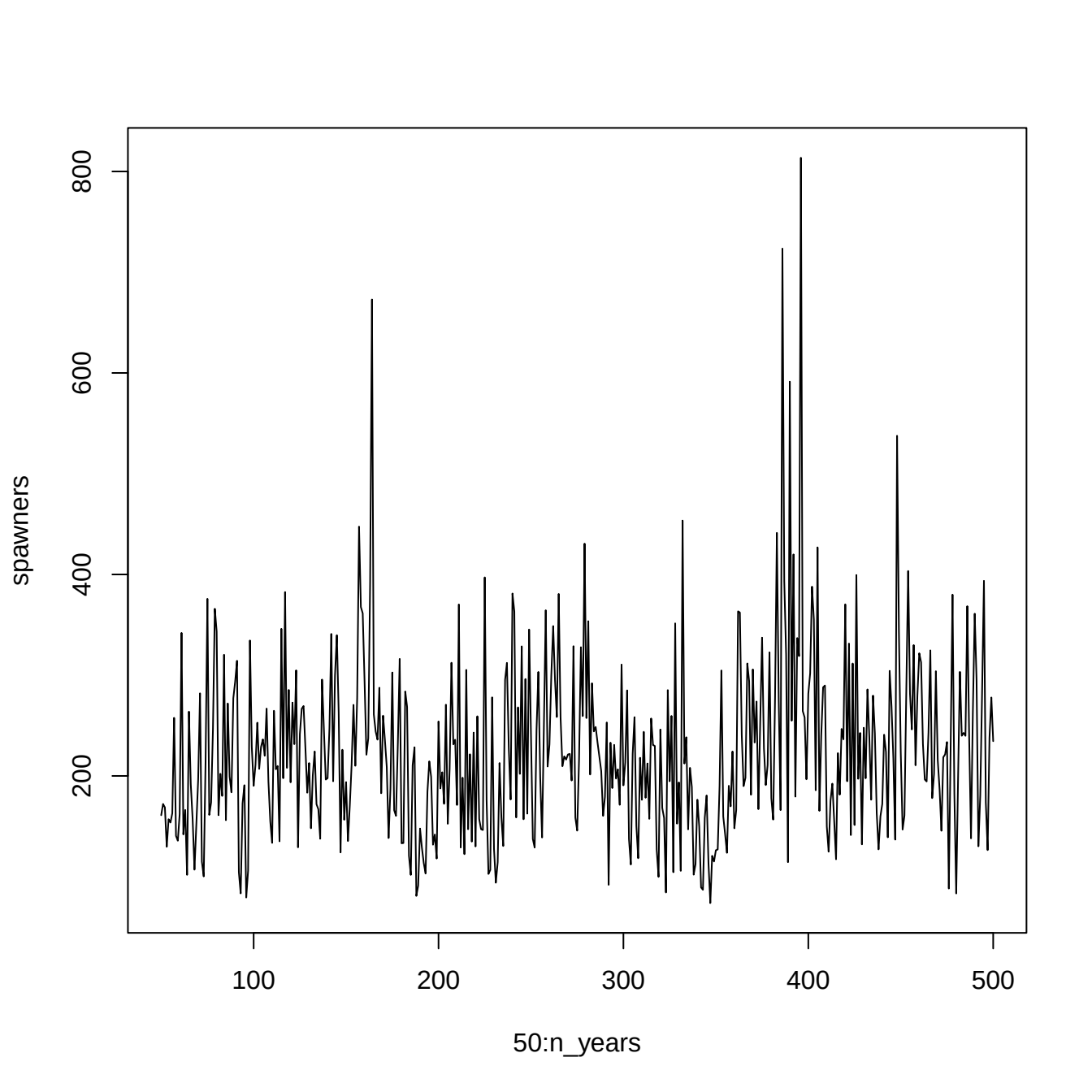

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

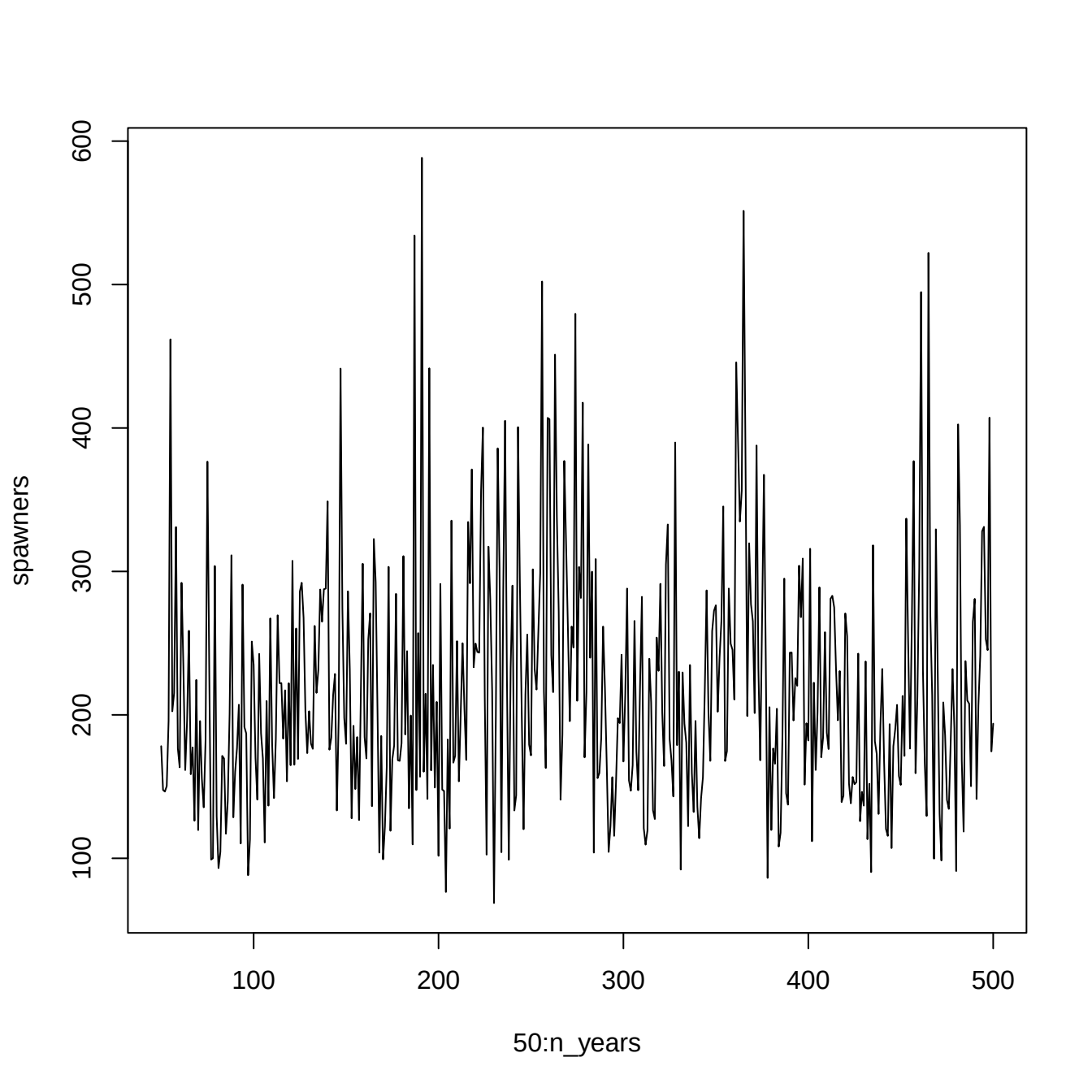

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

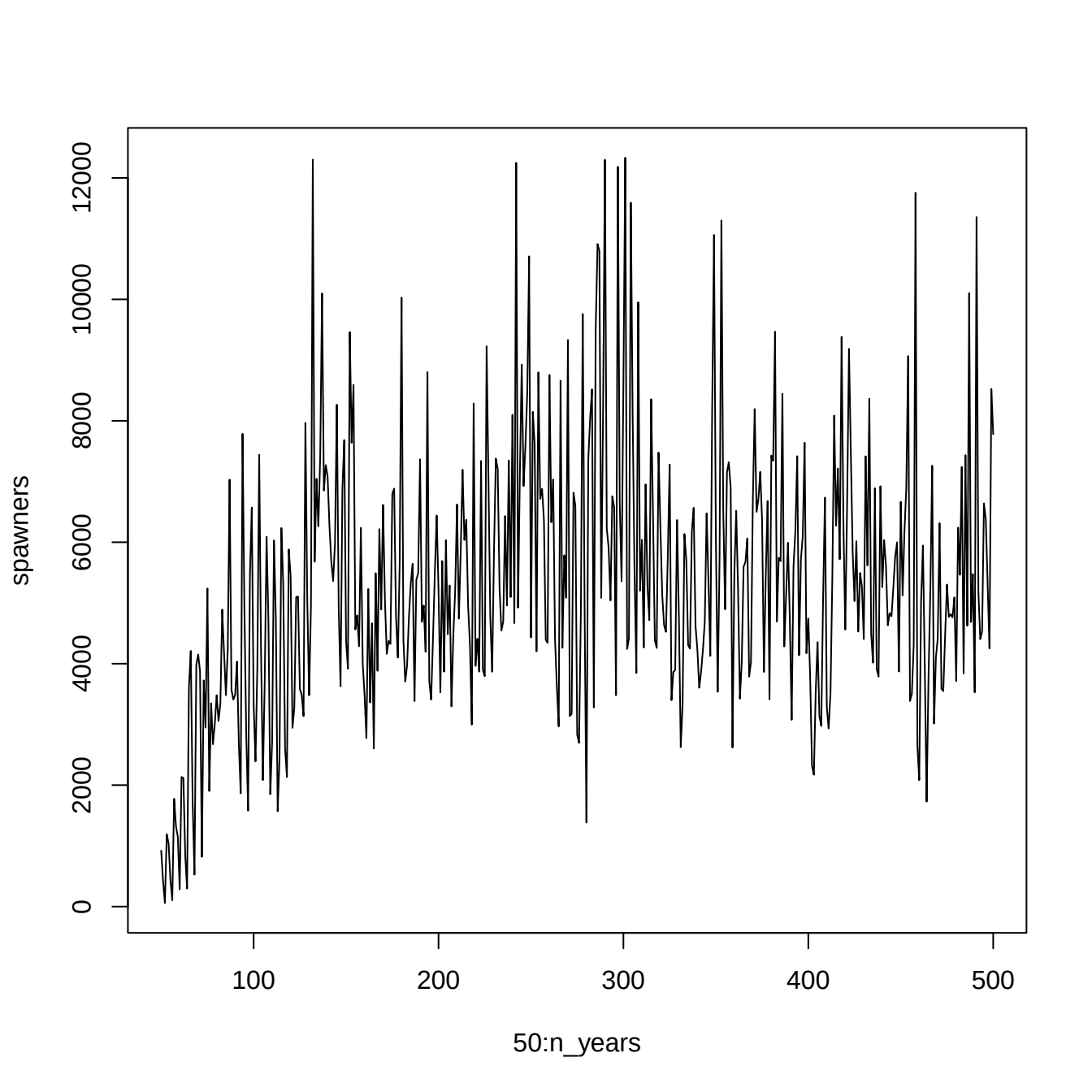

baselinepopN[50:200]
0
1000
2000
3000
4000
5000
0
50
100
150
Index

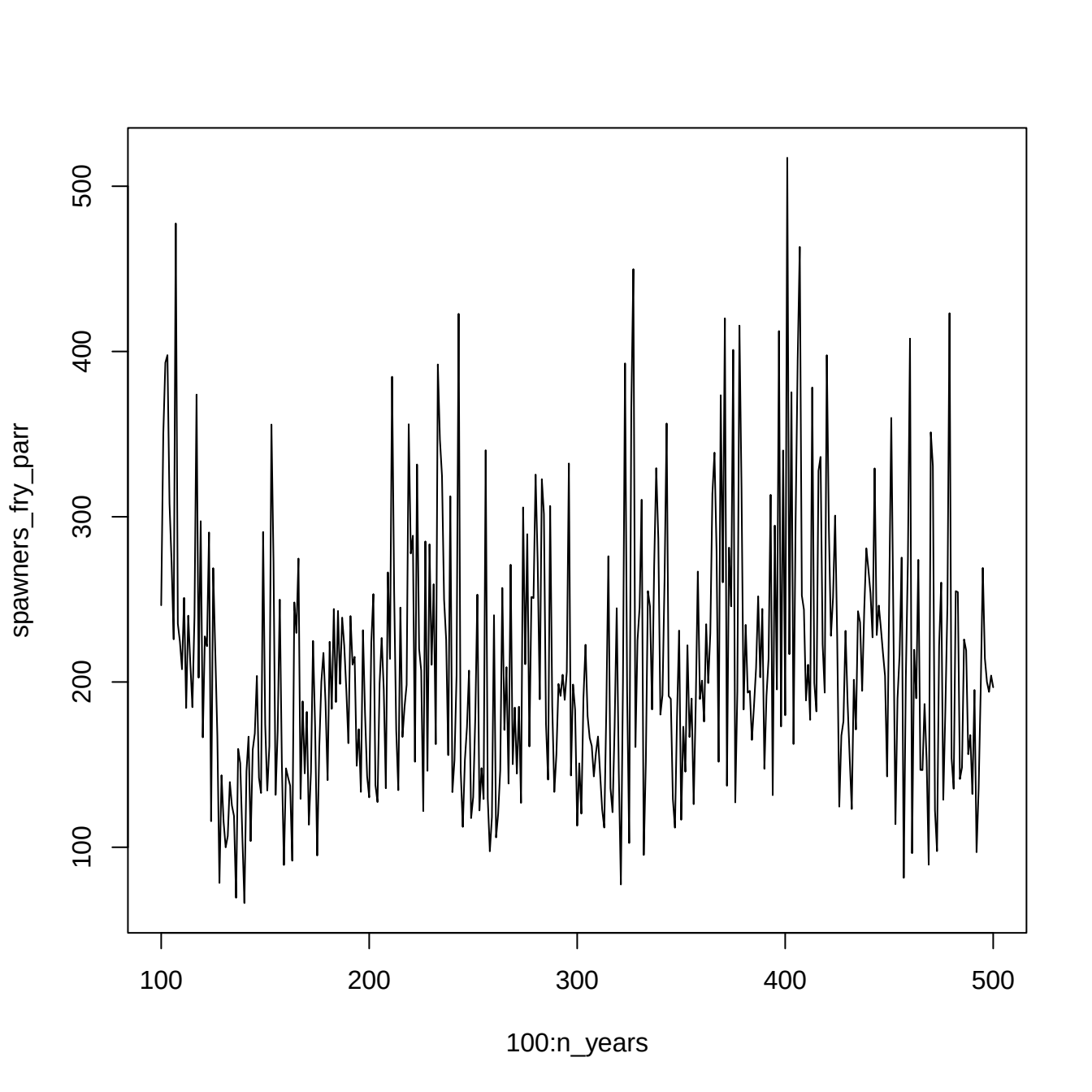

df$N
0e+00
2e−11
4e−11
6e−11
8e−11
100
200
300
400
500
df$year

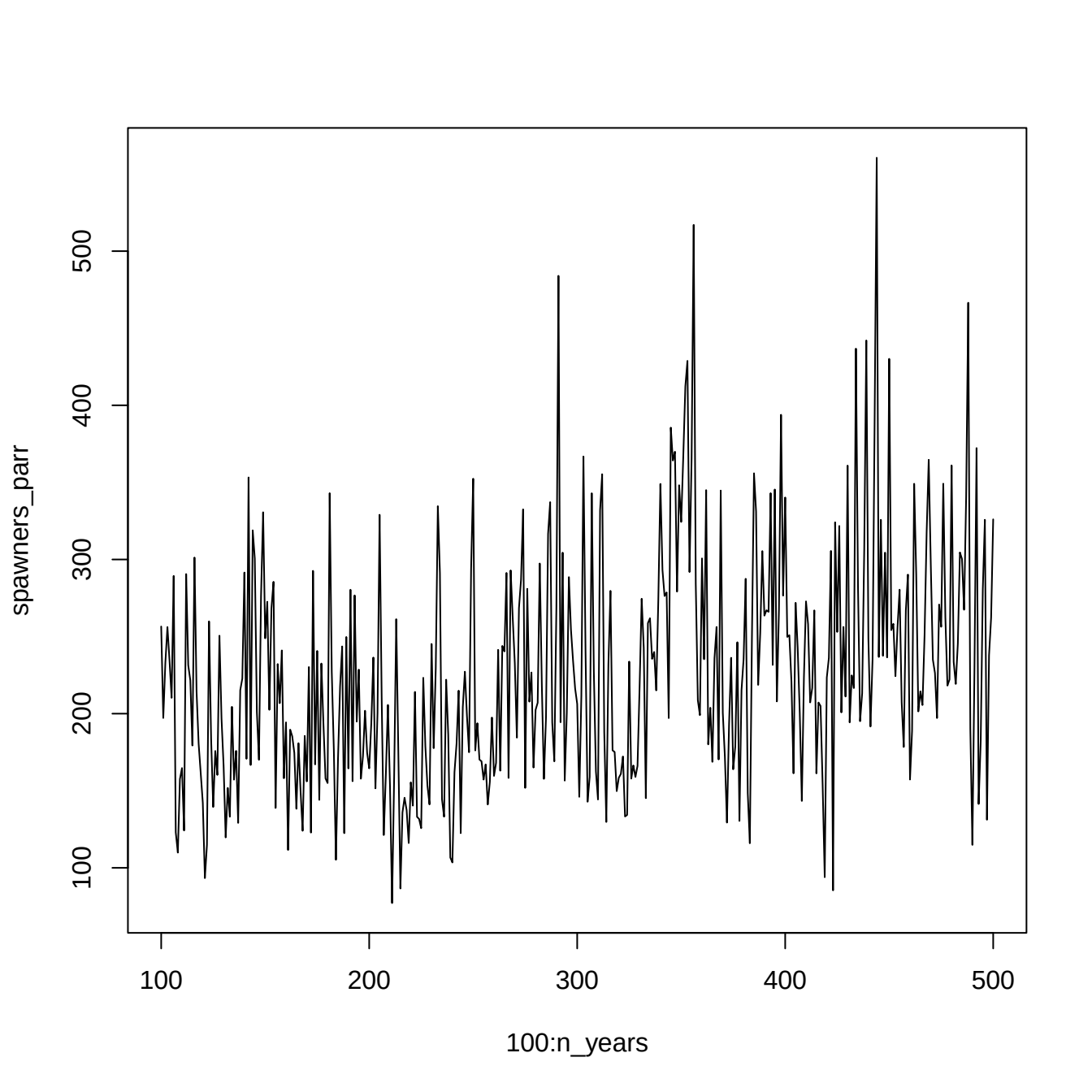

spawners
2000
2500
3000
3500
4000
4500
5000
100
200
300
400
500
50:n_years

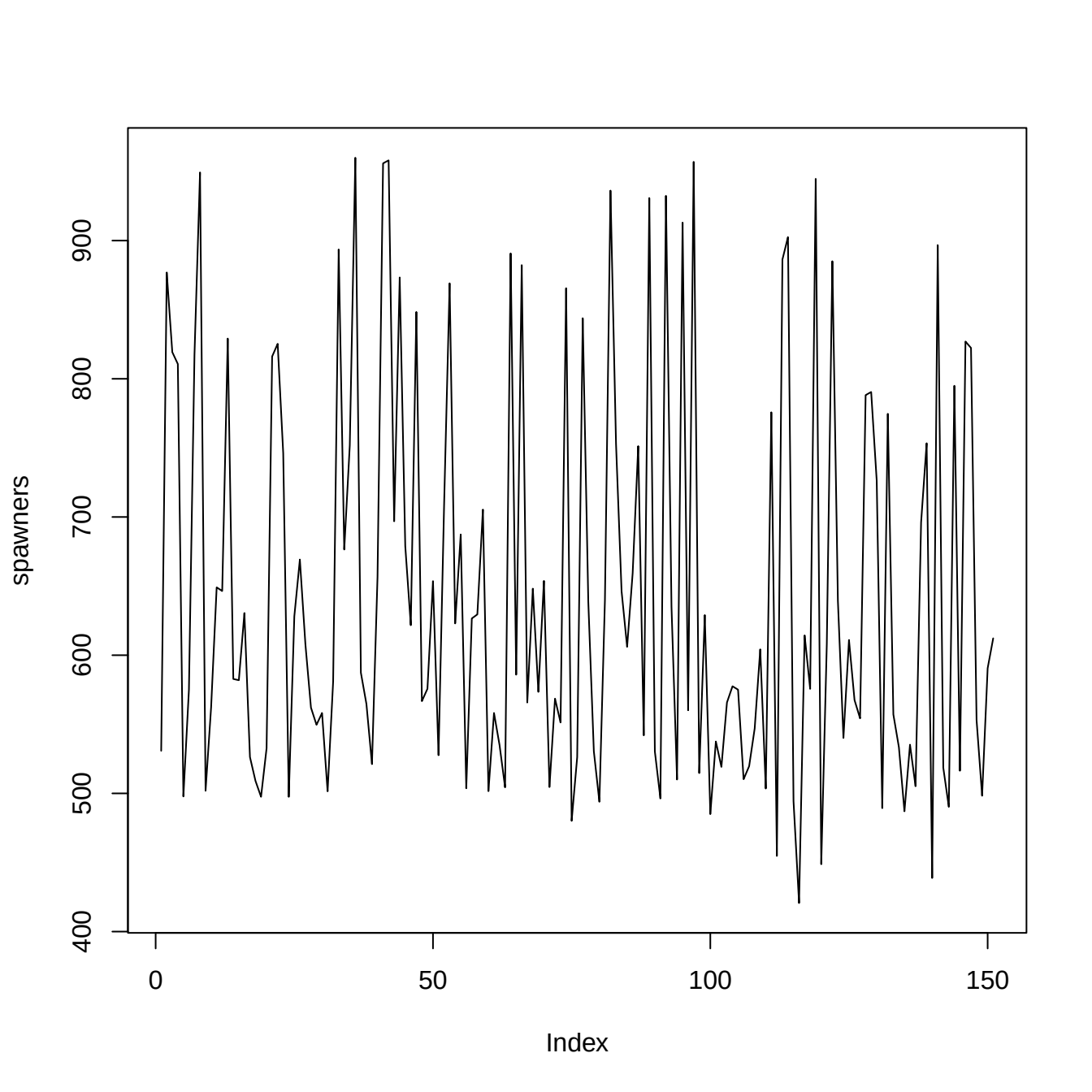

spawners
800
600
400
200
100
200
300
400
500
50:n_years

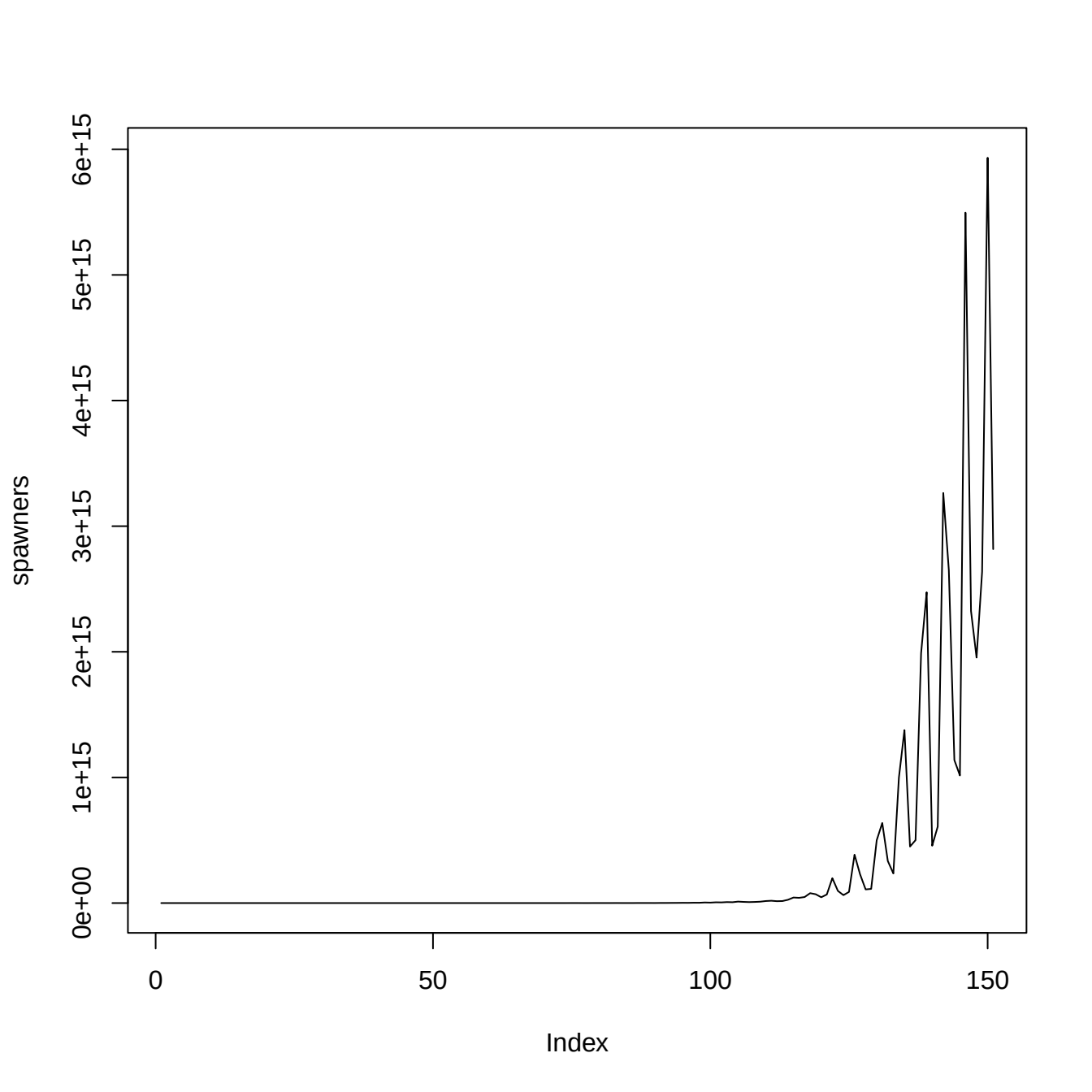

spawners
100
200
300
400
500
600
50:n_years
100
200
300
400
500

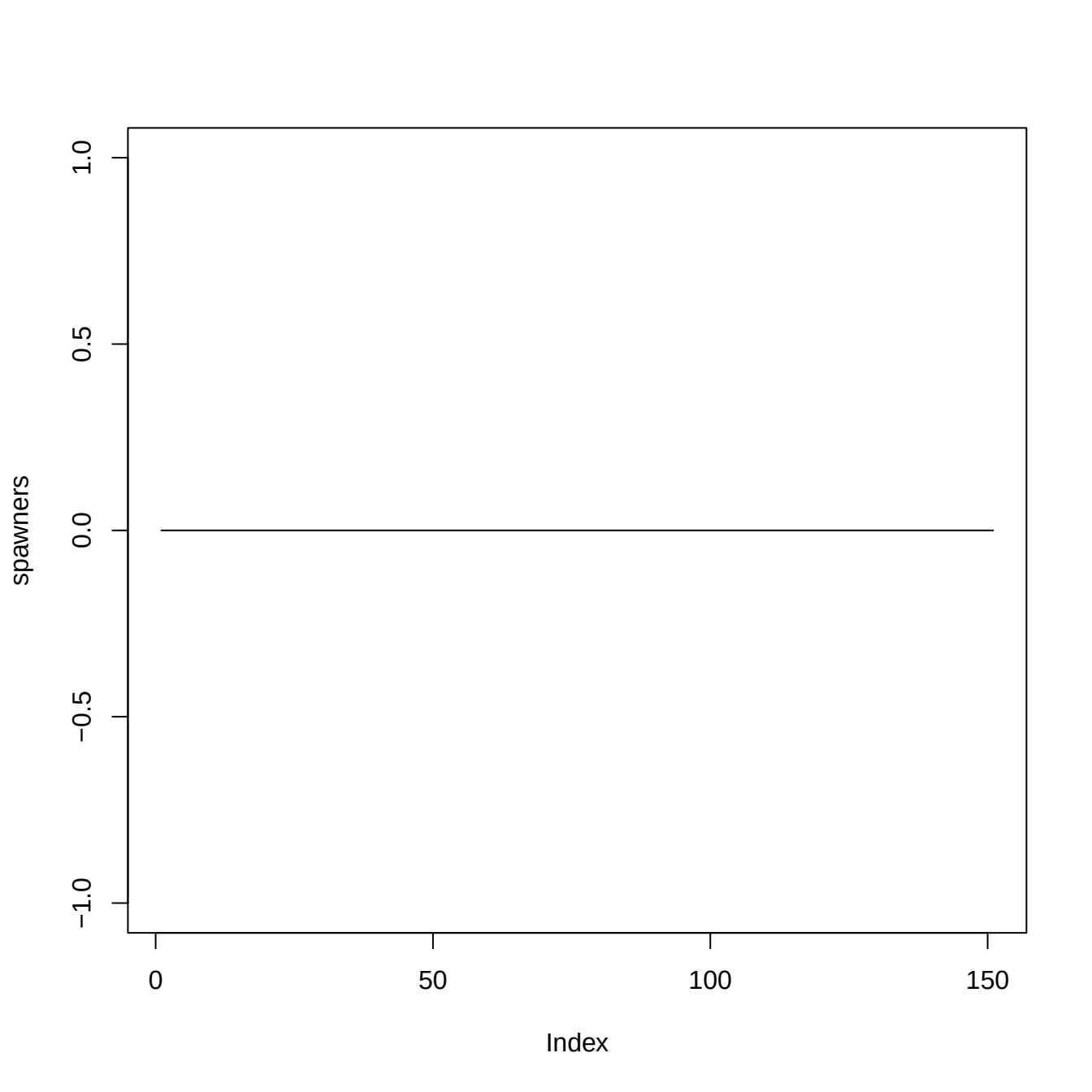

spawners
0
2000
4000
6000
8000
10000
12000
100
200
300
400
500
50:n_years

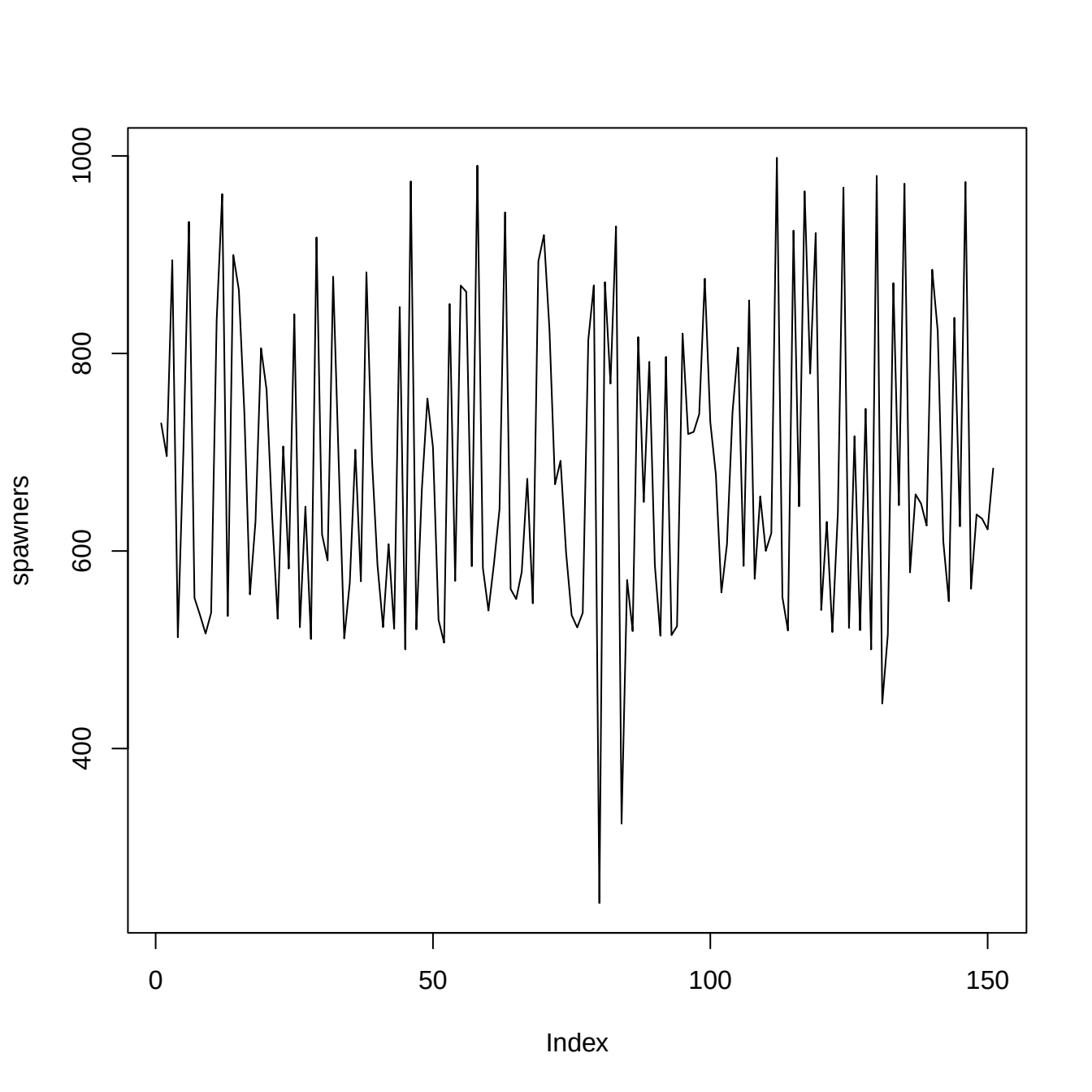

spawners_fry_parr
100
200
300
400
500
100
200
300
400
500
100:n_years

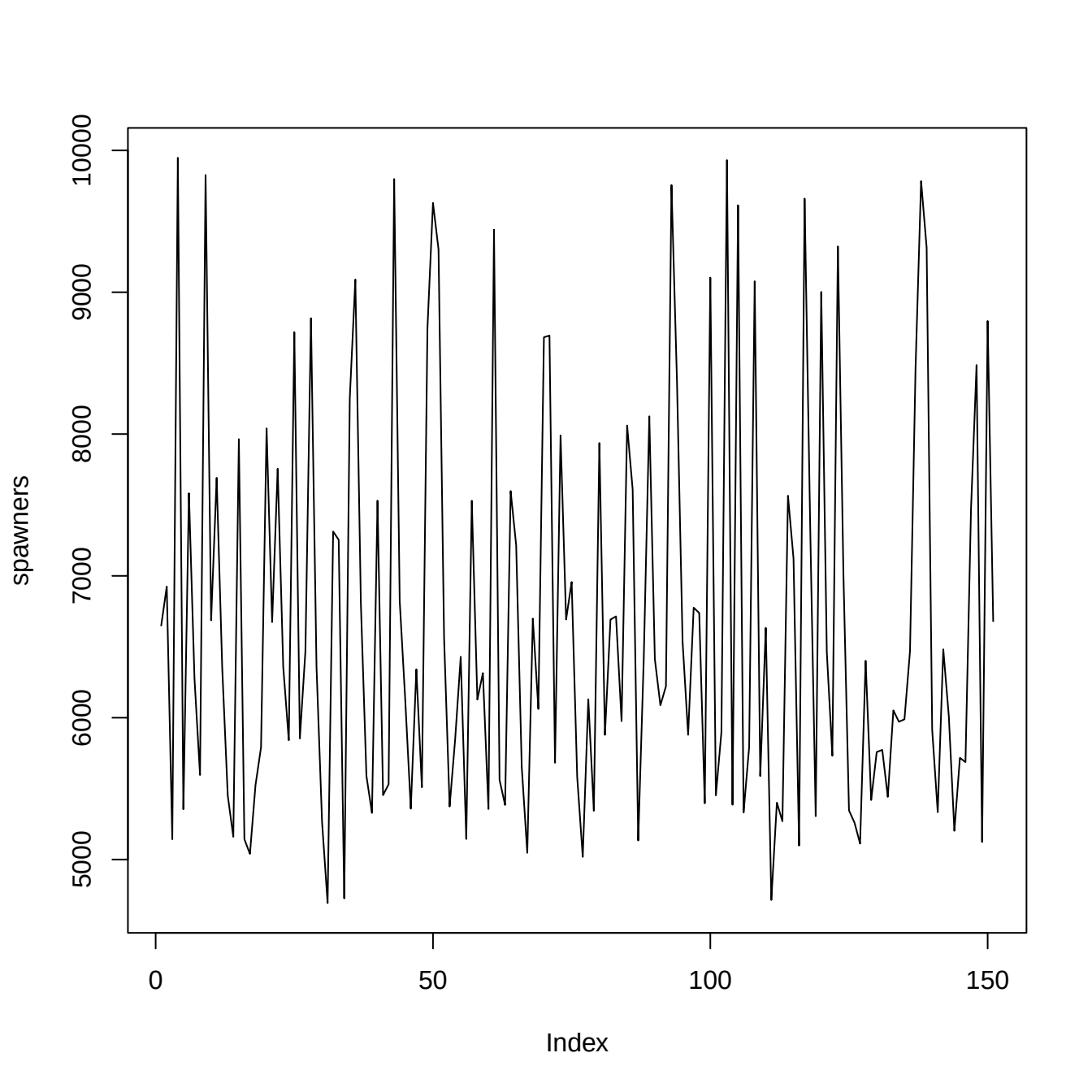

spawners_parr
100
200
300
400
500
100
200
300
400
500
100:n_years

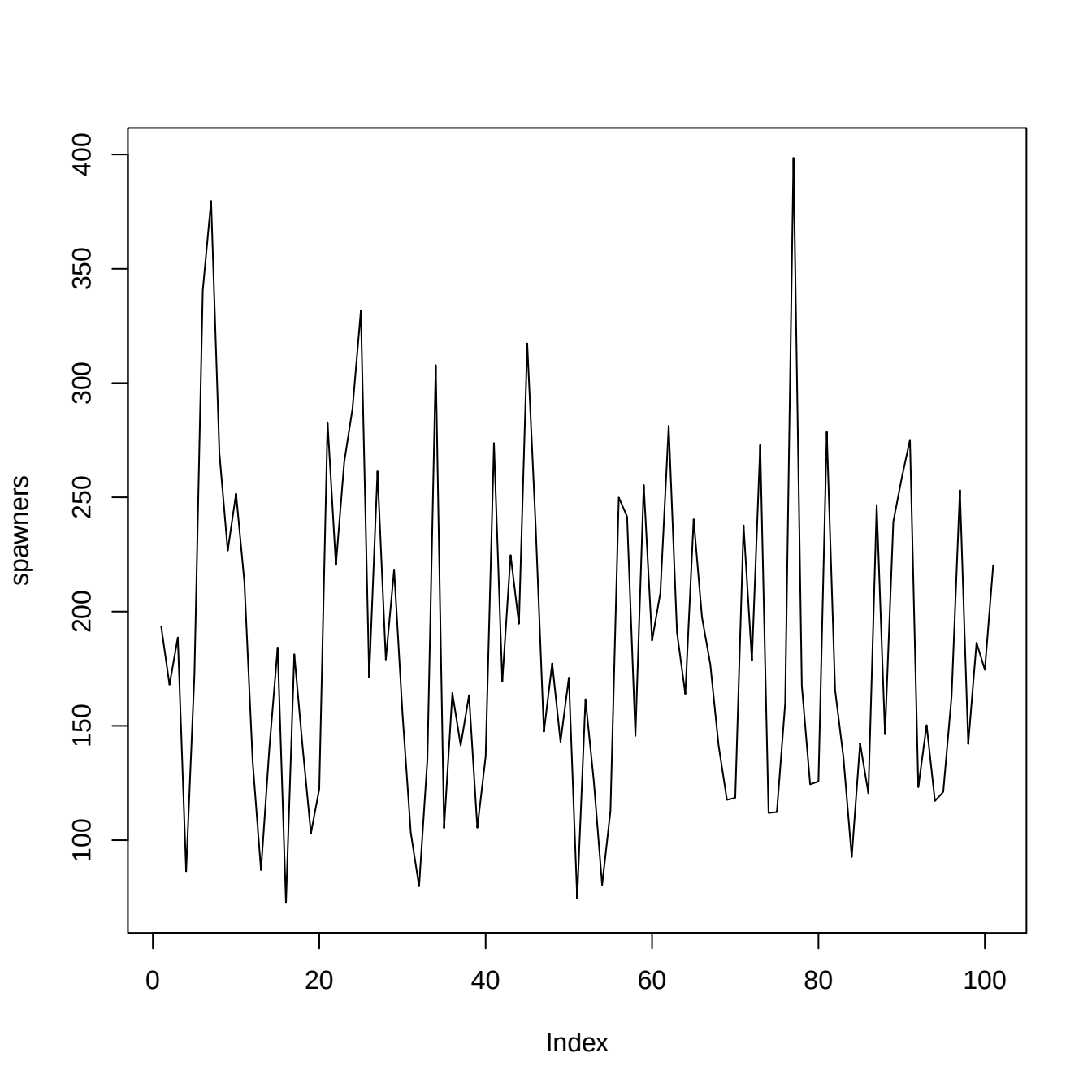

spawners
400
500
600
700
800
900
0
50
100
150
Index

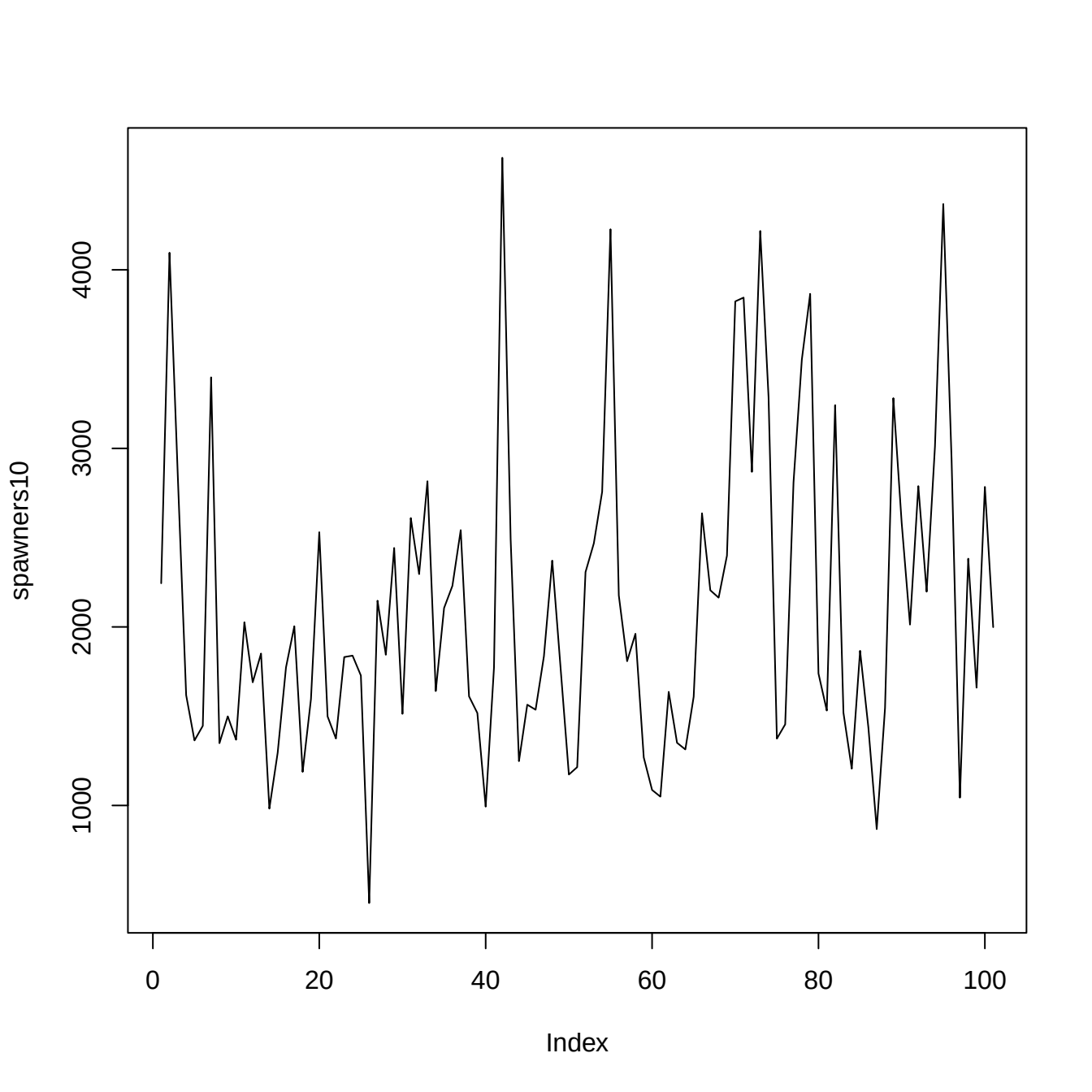

spawners
0e+00
1e+15
2e+15
3e+15
4e+15
5e+15
6e+15
0
50
100
150
Index

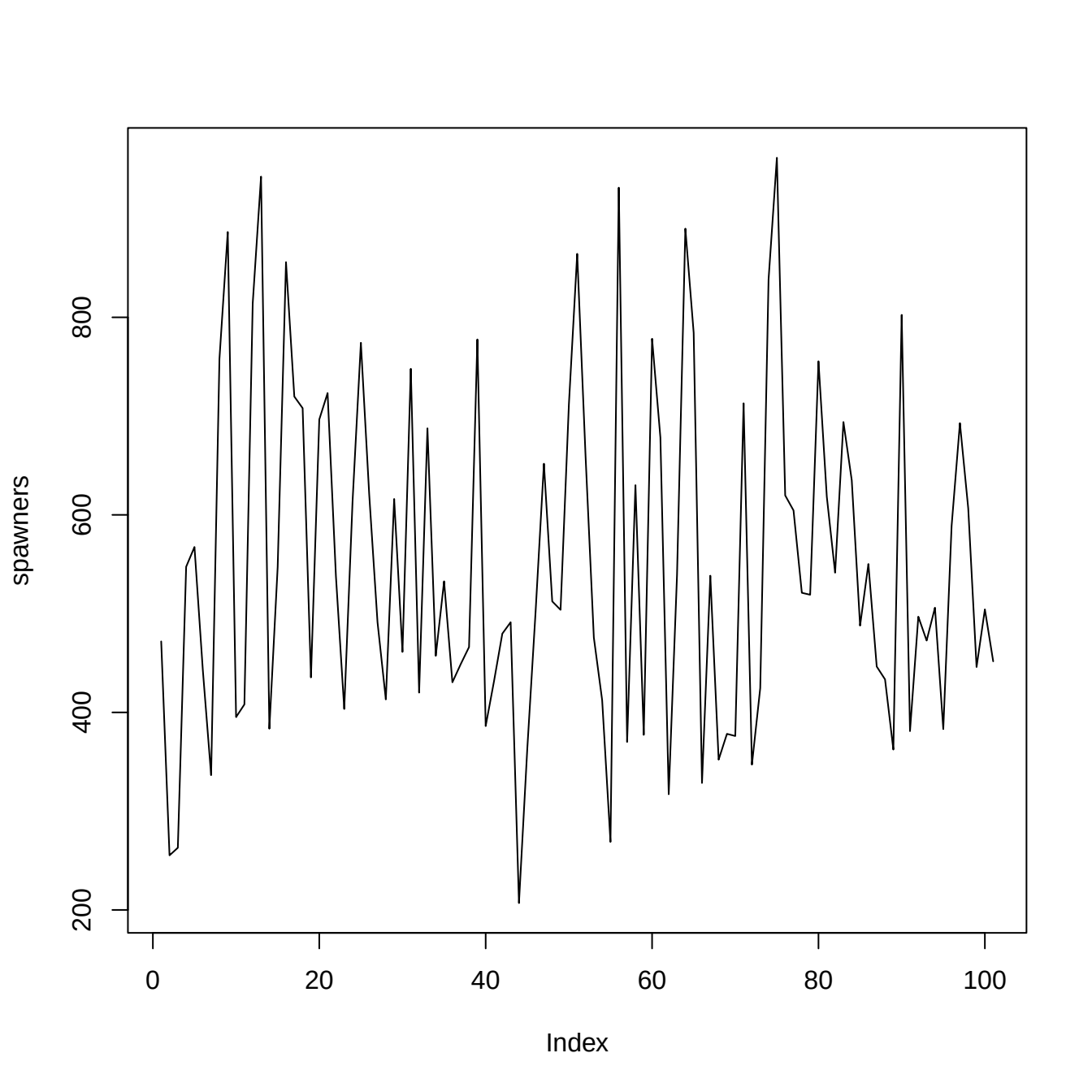

spawners
1.0
0.5
0.0
−0.5
−1.0
0
50
100
150
Index

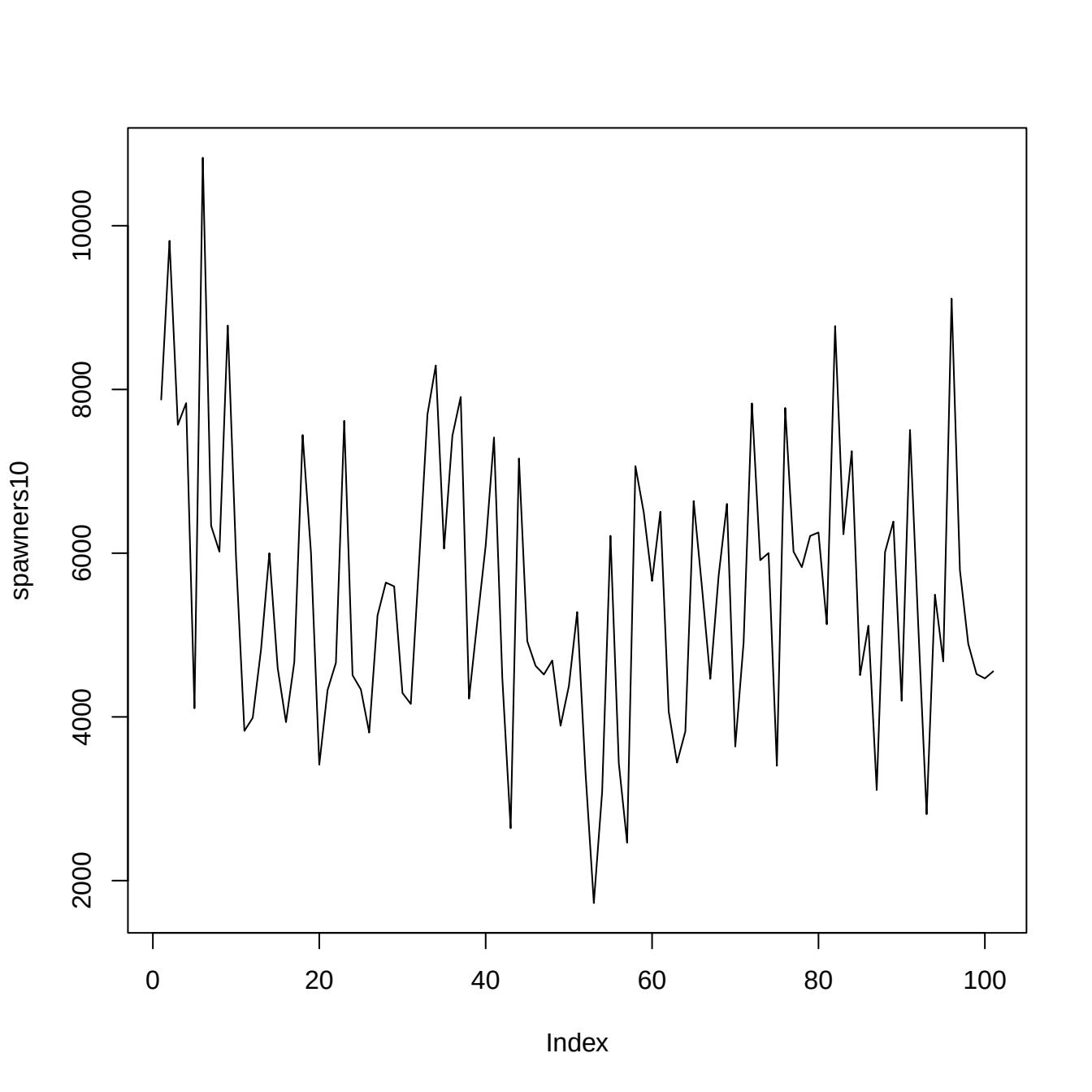

spawners
1000
800
600
400
0
50
100
150
Index

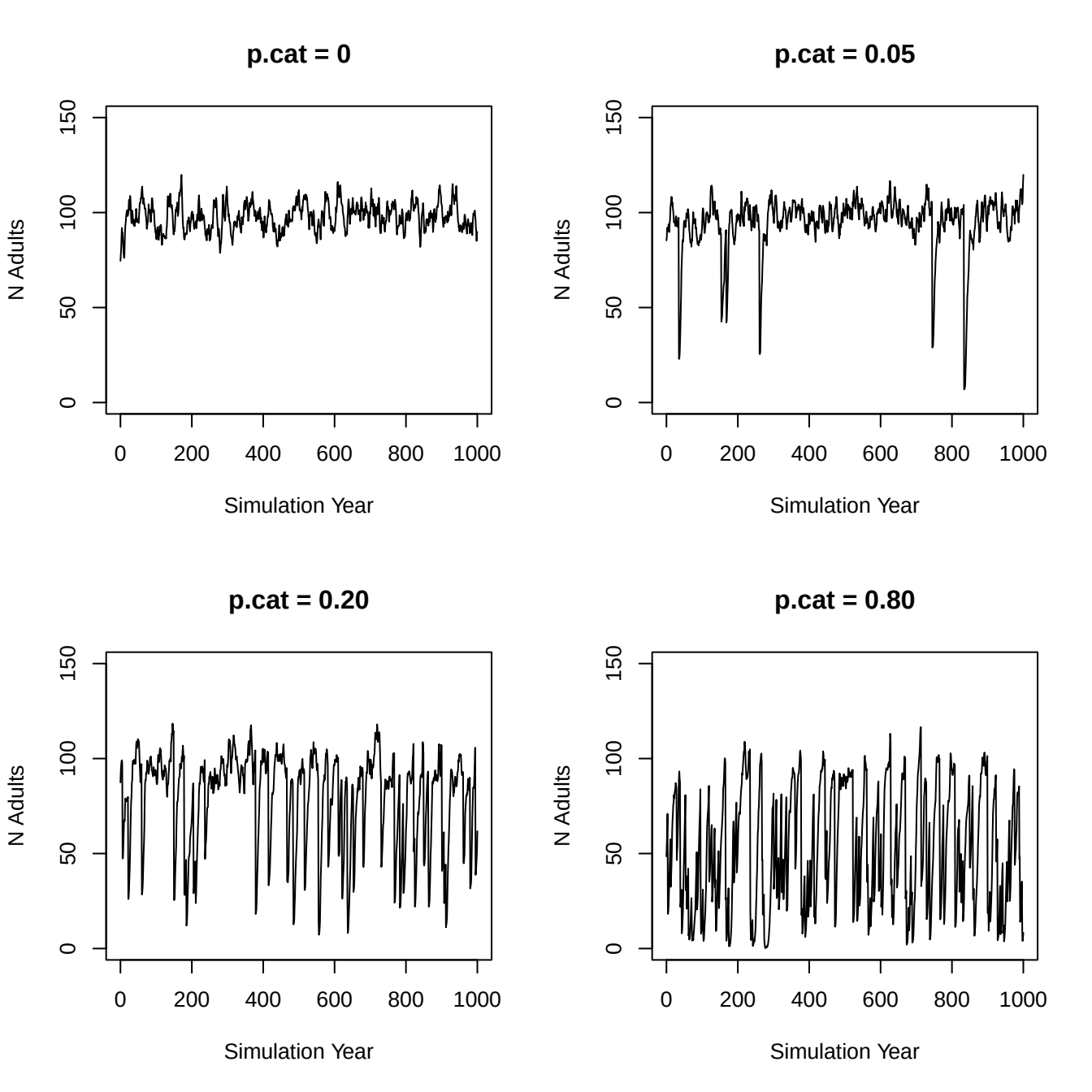

spawners
10000
9000
8000
7000
6000
5000
0
50
100
150
Index

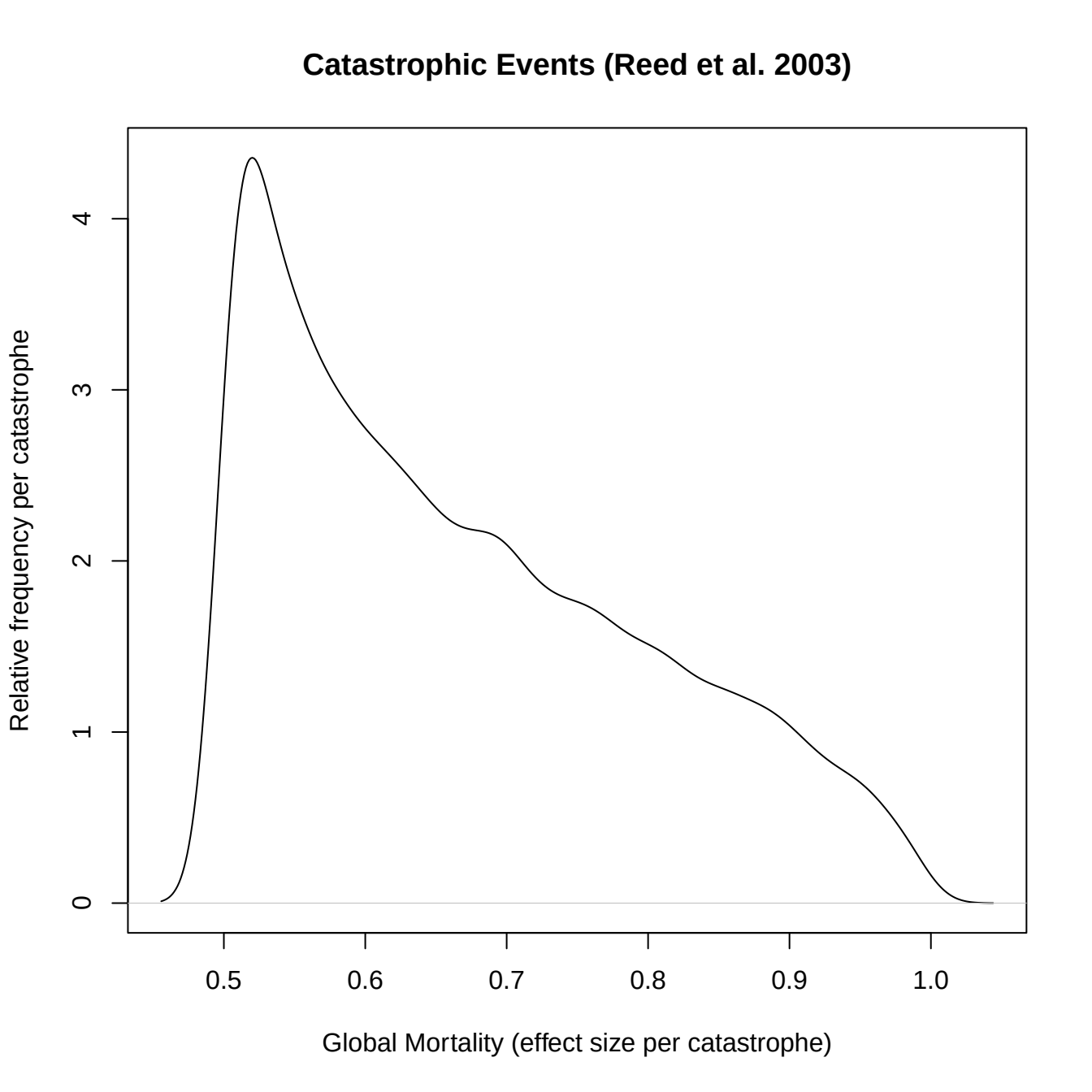

spawners
100
150
200
250
300
350
400
0
20
40
60
80
100
Index

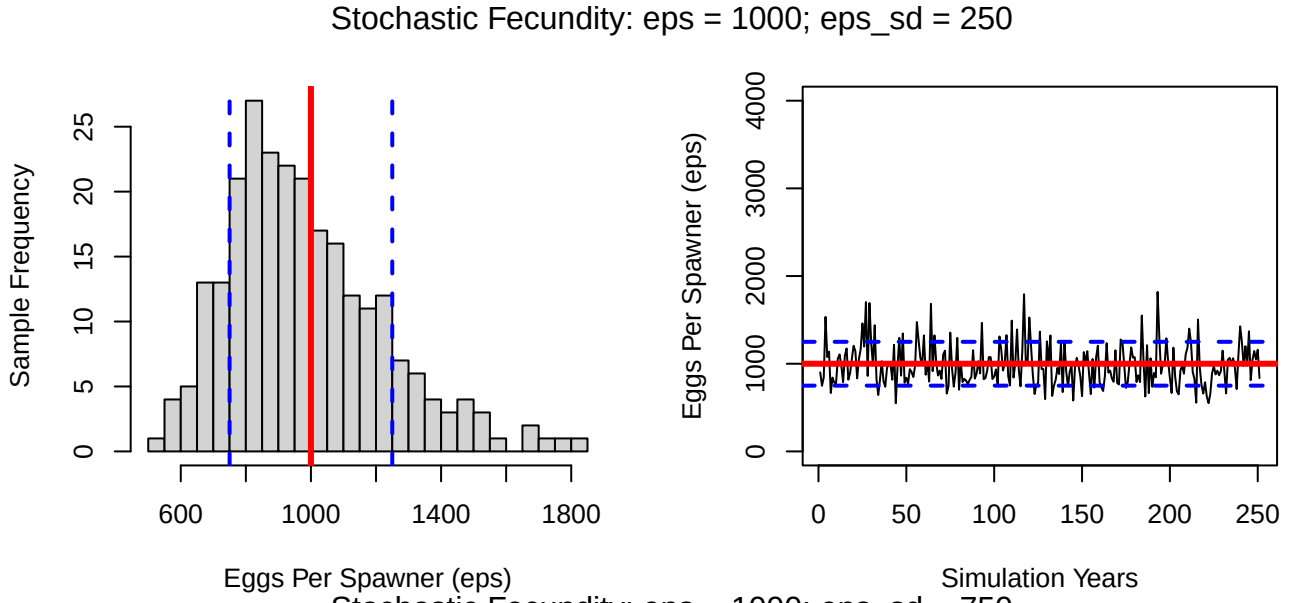

spawners10
1000
2000
3000
4000
0
20
40
60
80
100
Index

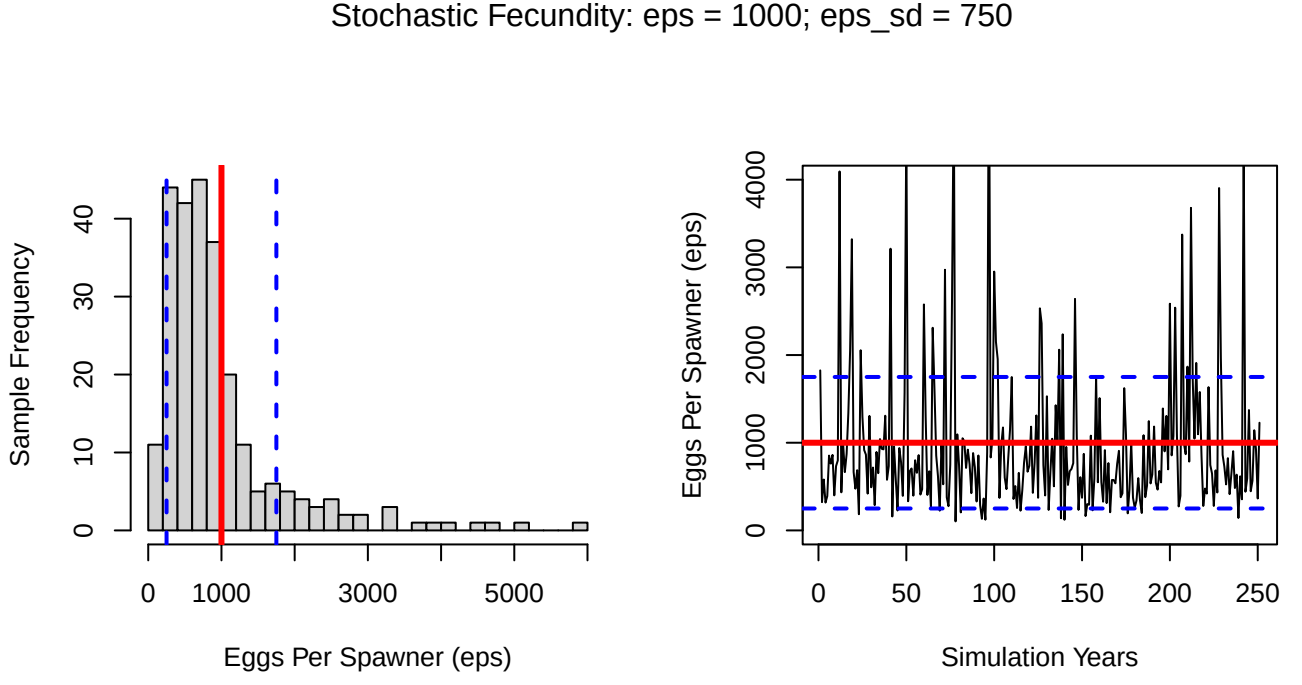

spawners
200
400
600
800
0
20
40
60
80
100
Index


spawners10
2000
4000
6000
8000
10000
0
20
40
60
80
100
Index

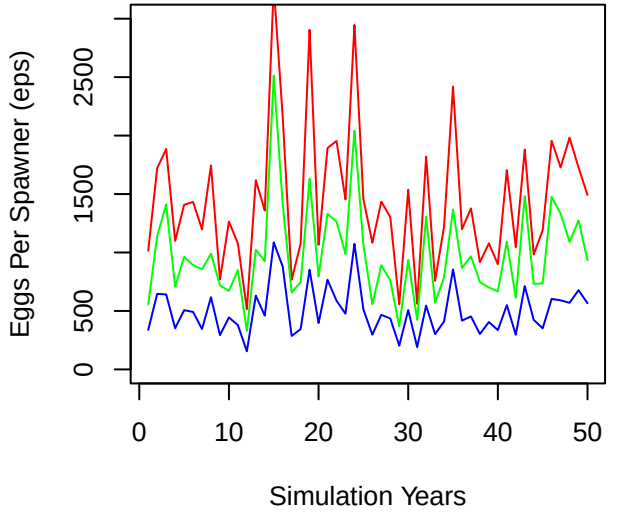

p.cat = 0
p.cat = 0.05
p.cat = 0.20
p.cat = 0.80
N Adults
Simulation Year
0
50
100
150
0
200
400
600
800
1000

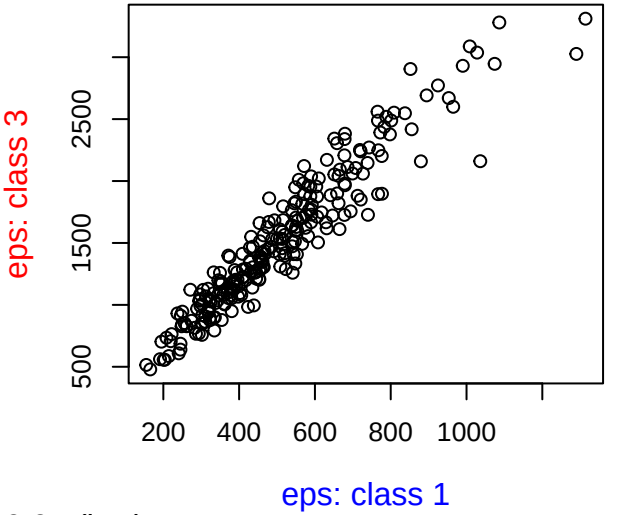

Catastrophic Events (Reed et al. 2003)
Relative frequency per catastrophe
0
1
2
3
4
0.5
0.6
0.7
0.8
0.9
1.0
Global Mortality (effect size per catastrophe)


Stochastic Fecundity: eps = 1000; eps_sd = 250
Sample Frequency
0
5
10
15
20
25
600
1000
1400
1800
Eggs Per Spawner (eps)
Eggs Per Spawner (eps)
0
1000
2000
3000
4000
0
50
100
150
200
250
Simulation Years

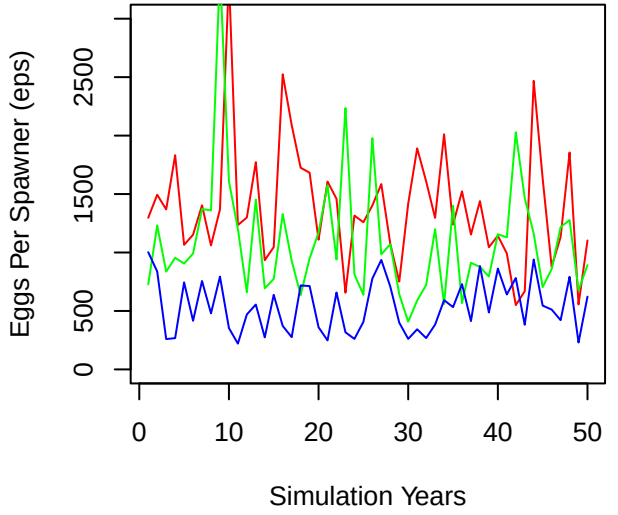

Stochastic Fecundity: eps = 1000; eps_sd = 750
Sample Frequency
0
10
20
30
40
0
1000
3000
5000
Eggs Per Spawner (eps)
Eggs Per Spawner (eps)
0
1000
2000
3000
4000
0
50
100
150
200
250
Simulation Years

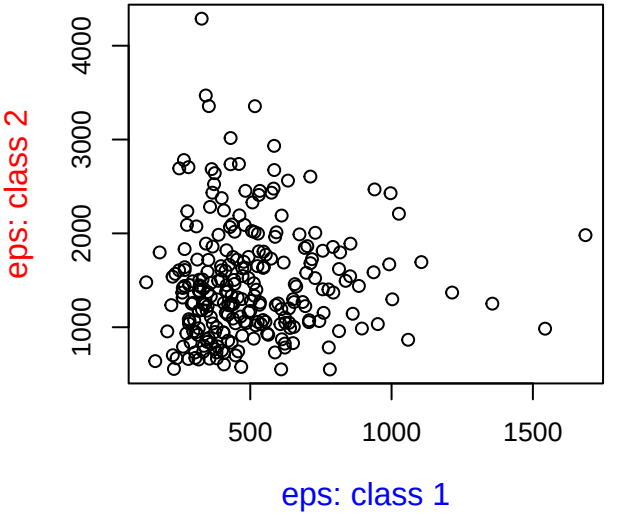
egg_rho = 0.95 (high)



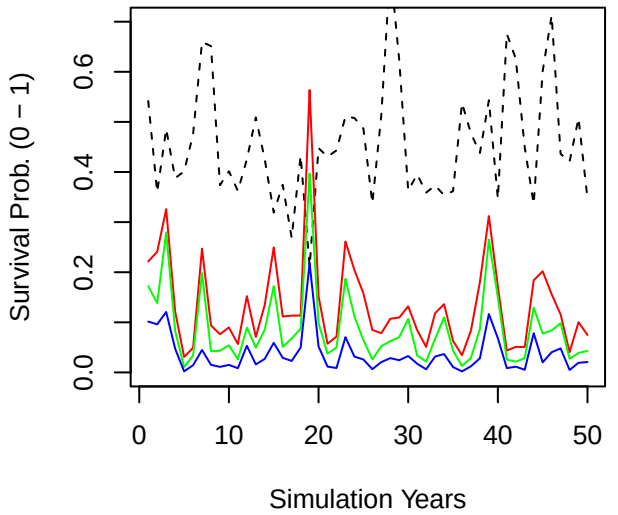


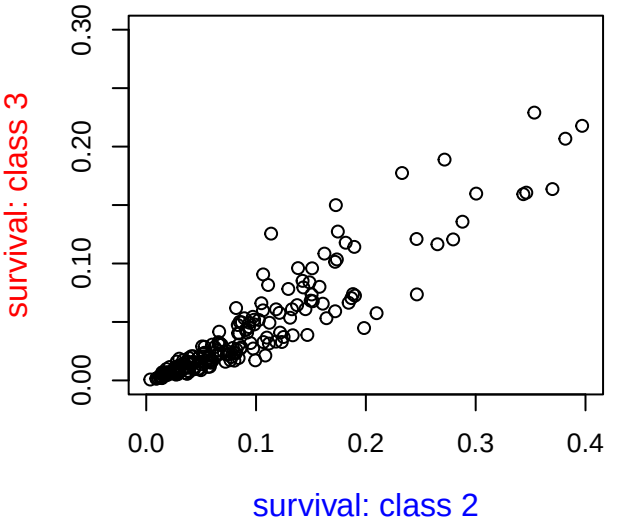
egg_rho = 0.05 (low)



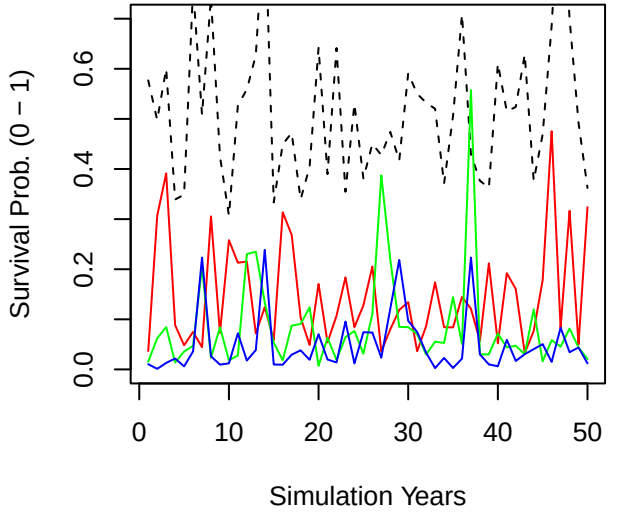


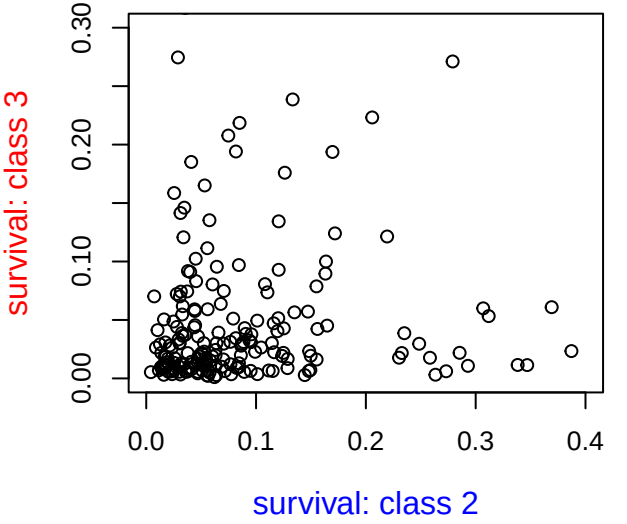
M.rho = 0.95 (high)

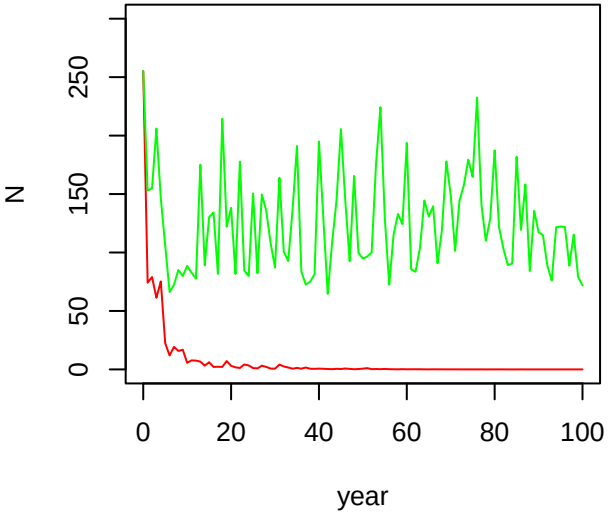


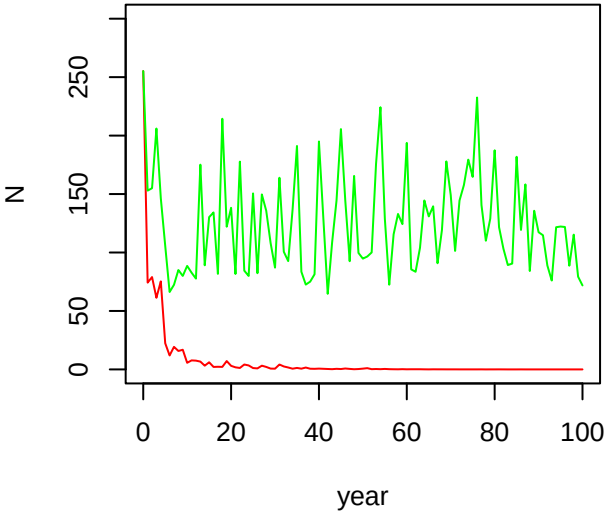


M.rho = 0.05 (low)






N
0
50
150
250
0
20
40
60
80
100
year


N
0
50
150
250
0
20
40
60
80
100
year